US 20250285058A1

(19) United States
(12) Patent Application Publication
Woods et al.

(10) Pub. No.: US 2025/0285058 A1
(43) Pub. Date: Sep. 11, 2025

(54) **SYSTEMS, METHODS, AND SOFTWARE FOR ENHANCED RISK ASSESSMENT**

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(21) Appl. No.: **19/073,827**

(22) Filed: **Mar. 7, 2025**

Related U.S. Application Data

(60) Provisional application No. 63/562,958, filed on Mar. 8, 2024.

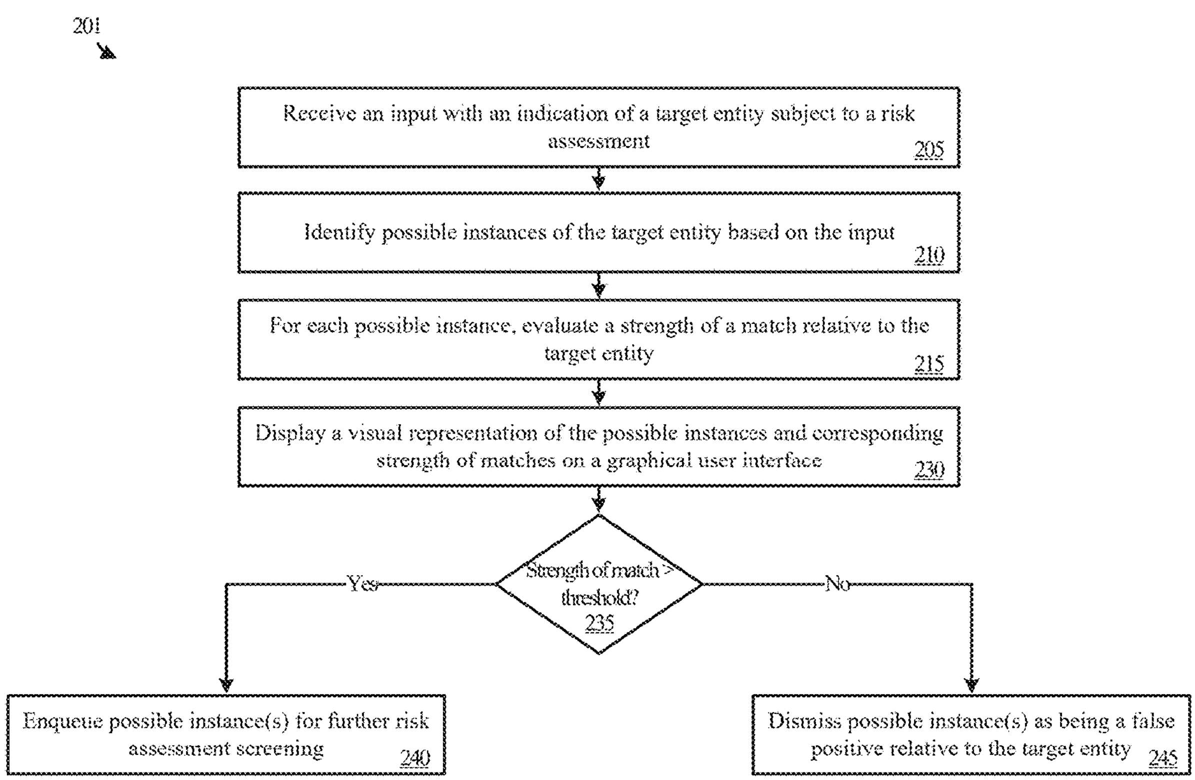
Publication Classification

(51) **Int. Cl.**
G06Q 10/0635 (2023.01)

(52) **U.S. Cl.**
CPC *G06Q 10/0635* (2013.01)

(57) **ABSTRACT**

Various examples disclosed herein relate to risk assessment and analysis of entities with respect to potential financial dealings. In an example embodiment, a risk assessment platform is provided that includes one or more computer-readable storage media and program instructions stored on the one or more computer-readable storage media that, based on being read and executed by a processing device, direct the processing device to perform various functions. Specifically, the program instructions direct the processing device to receive an input with an indication of a target entity, identify, based on the input, multiple possible instances of the target entity from a database, and for each possible instance of the multiple possible instances, evaluate a strength of a match of the possible instance to the target entity for display thereof on a graphical user interface.



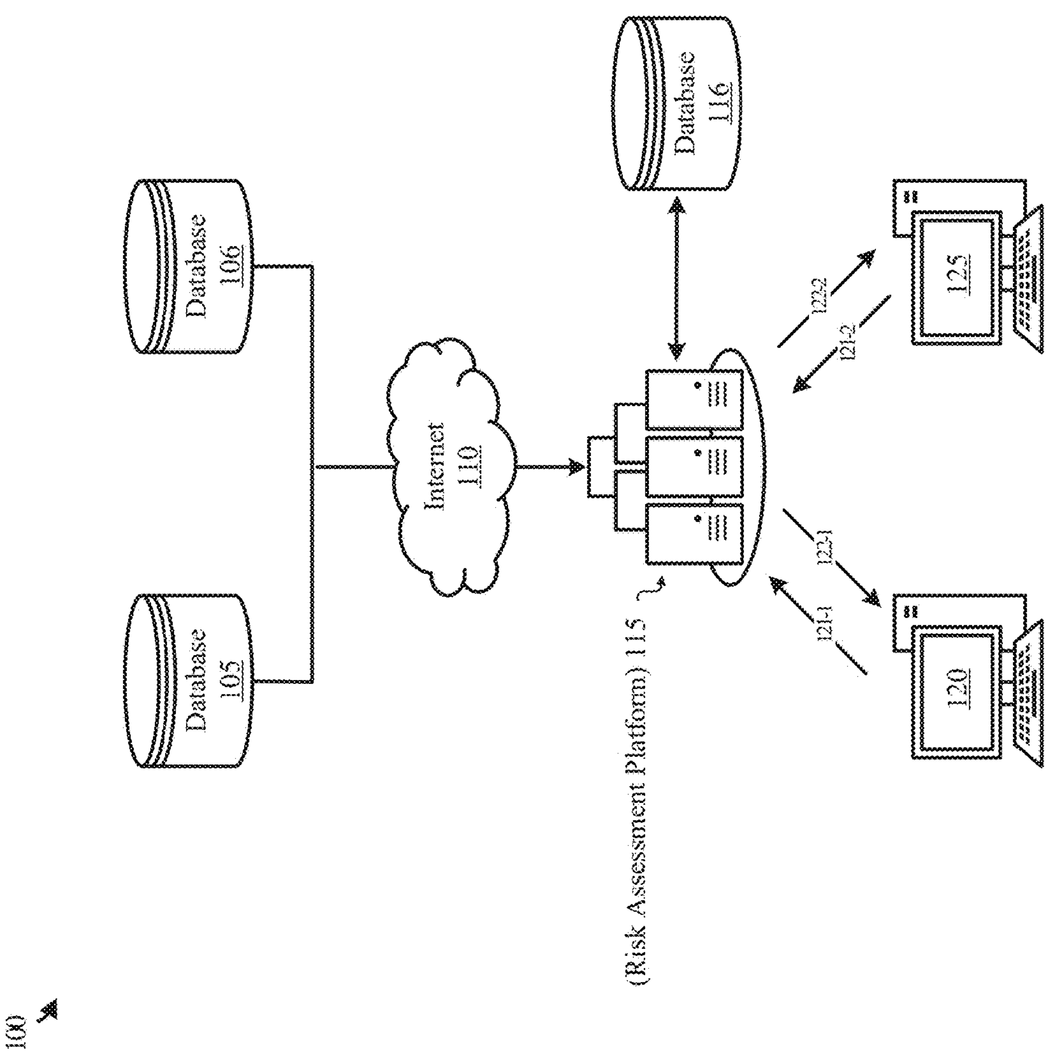

100
Database 105
Database 106
Internet 110
Database 116
(Risk Assessment Platform) 115
121-1
122-1
121-2
122-2
120
125

FIGURE 1

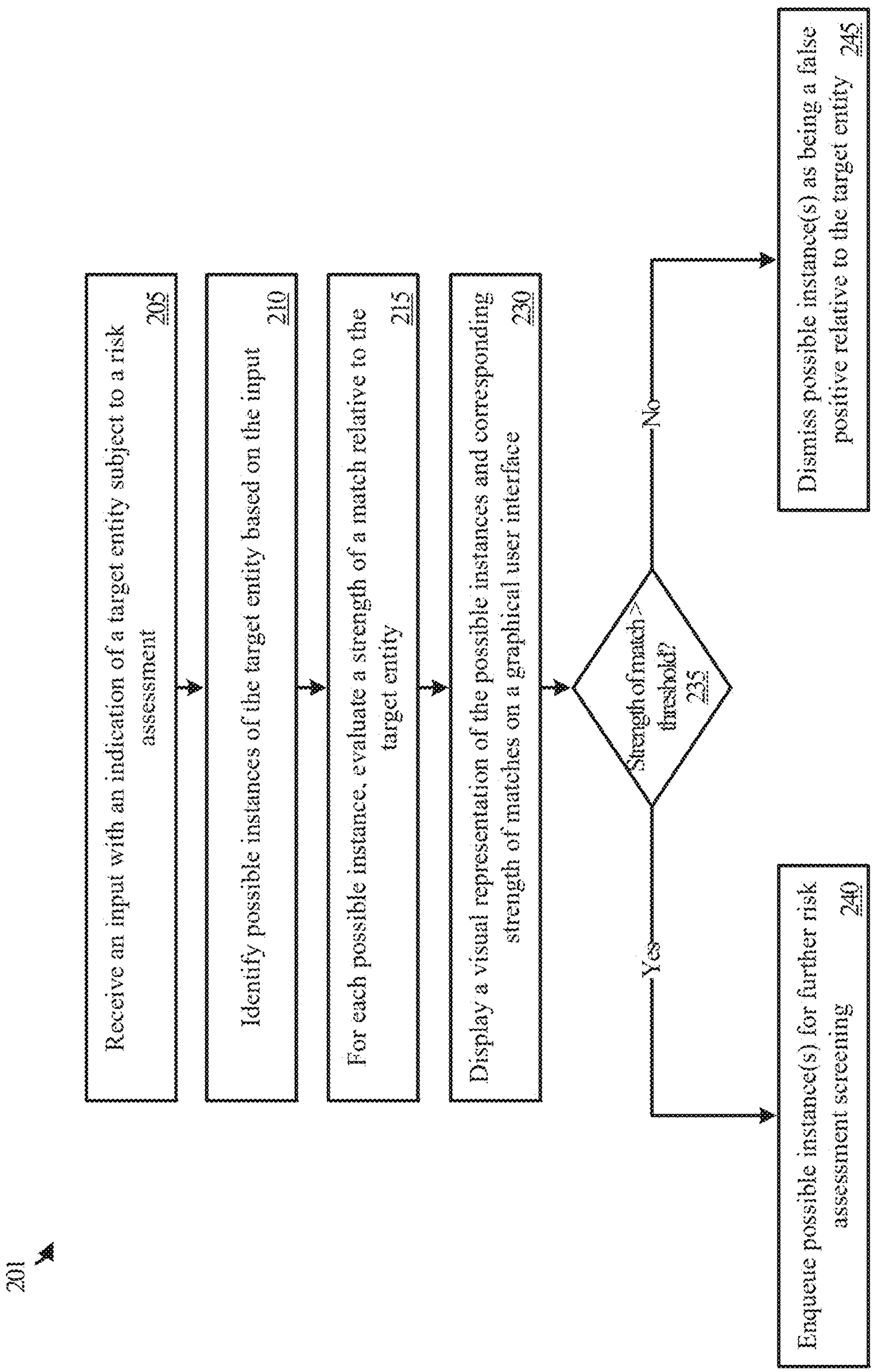

201
Receive an input with an indication of a target entity subject to a risk assessment 205
Identify possible instances of the target entity based on the input 210
For each possible instance, evaluate a strength of a match relative to the target entity 215
Display a visual representation of the possible instances and corresponding strength of matches on a graphical user interface 230
Strength of match > threshold? 235
Yes
No
Enqueue possible instance(s) for further risk assessment screening 240
Dismiss possible instance(s) as being a false positive relative to the target entity 245

FIGURE 2A

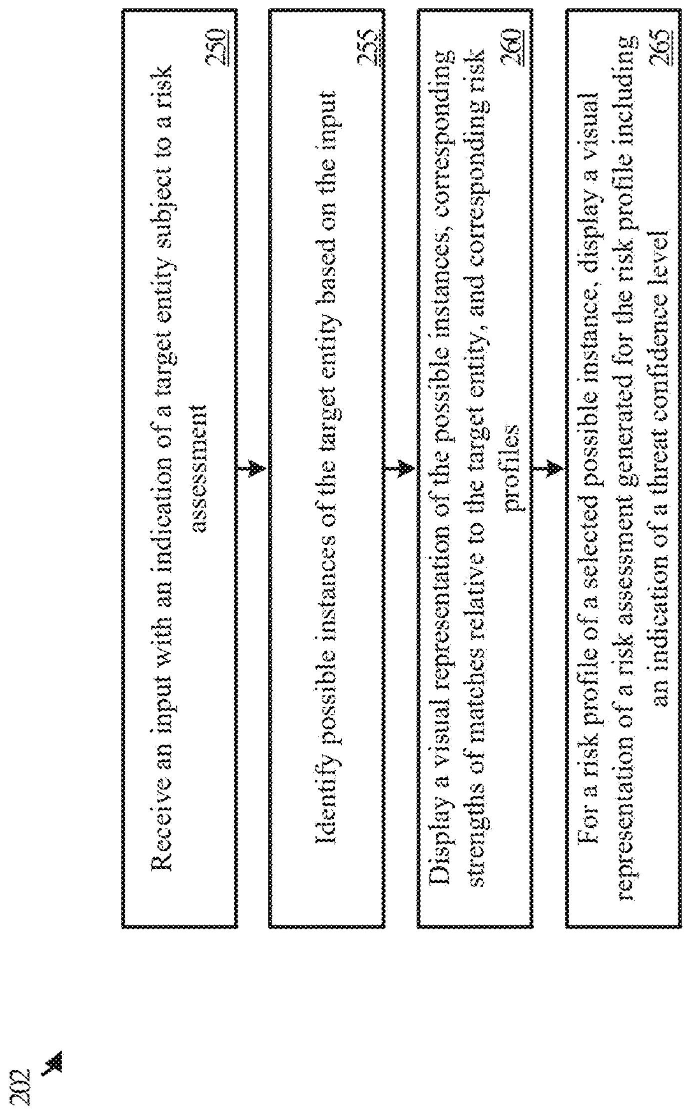

202
Receive an input with an indication of a target entity subject to a risk assessment
250
Identify possible instances of the target entity based on the input
255
Display a visual representation of the possible instances, corresponding strengths of matches relative to the target entity, and corresponding risk profiles
260
For a risk profile of a selected possible instance, display a visual representation of a risk assessment generated for the risk profile including an indication of a threat confidence level
265

FIGURE 2B

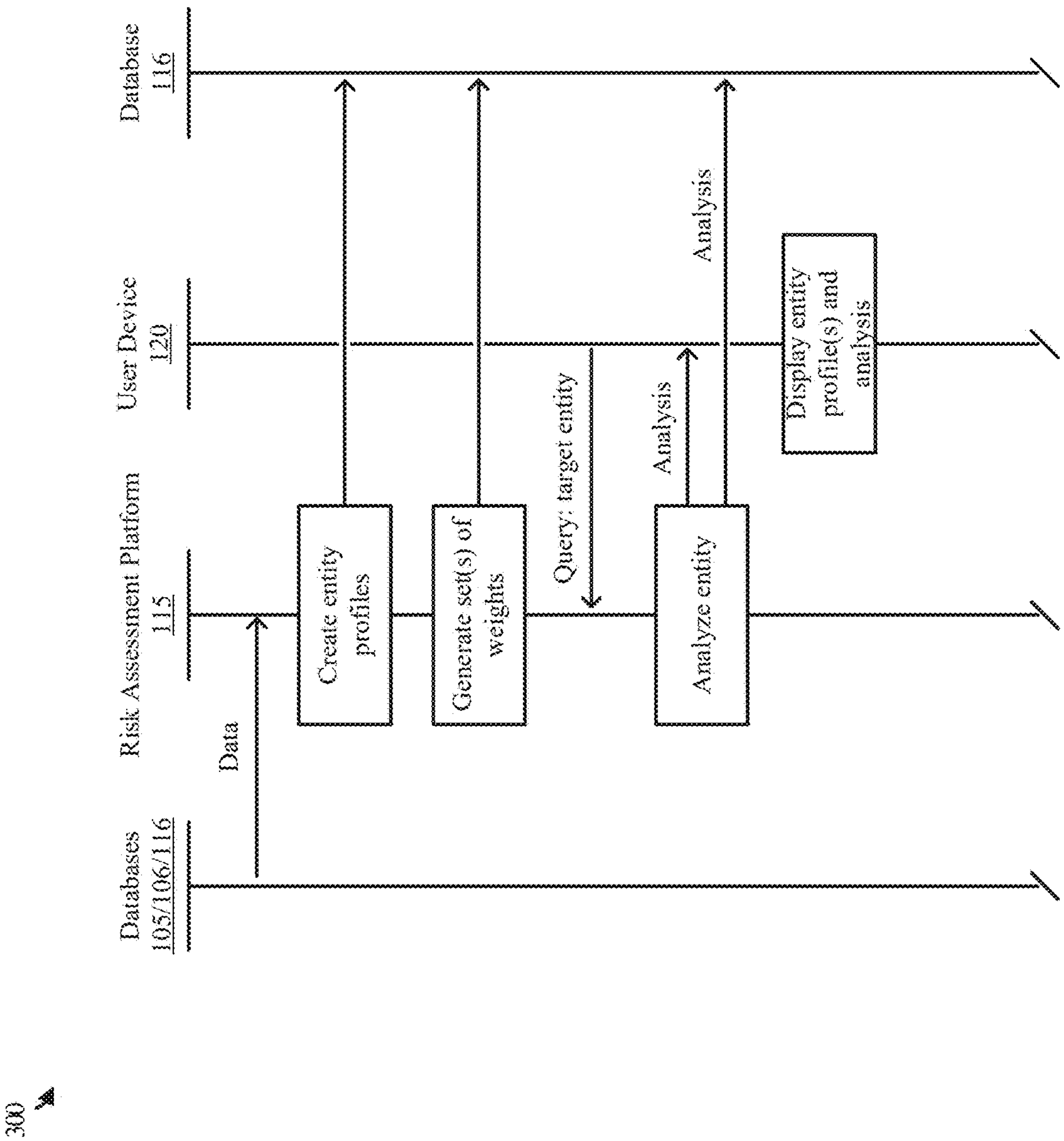

300
Databases
105/106/116
Risk Assessment Platform
115
User Device
120
Database
116
Data
Create entity profiles
Generate set(s) of weights
Query: target entity
Analyze entity
Analysis
Analysis
Display entity profile(s) and analysis

FIGURE 3

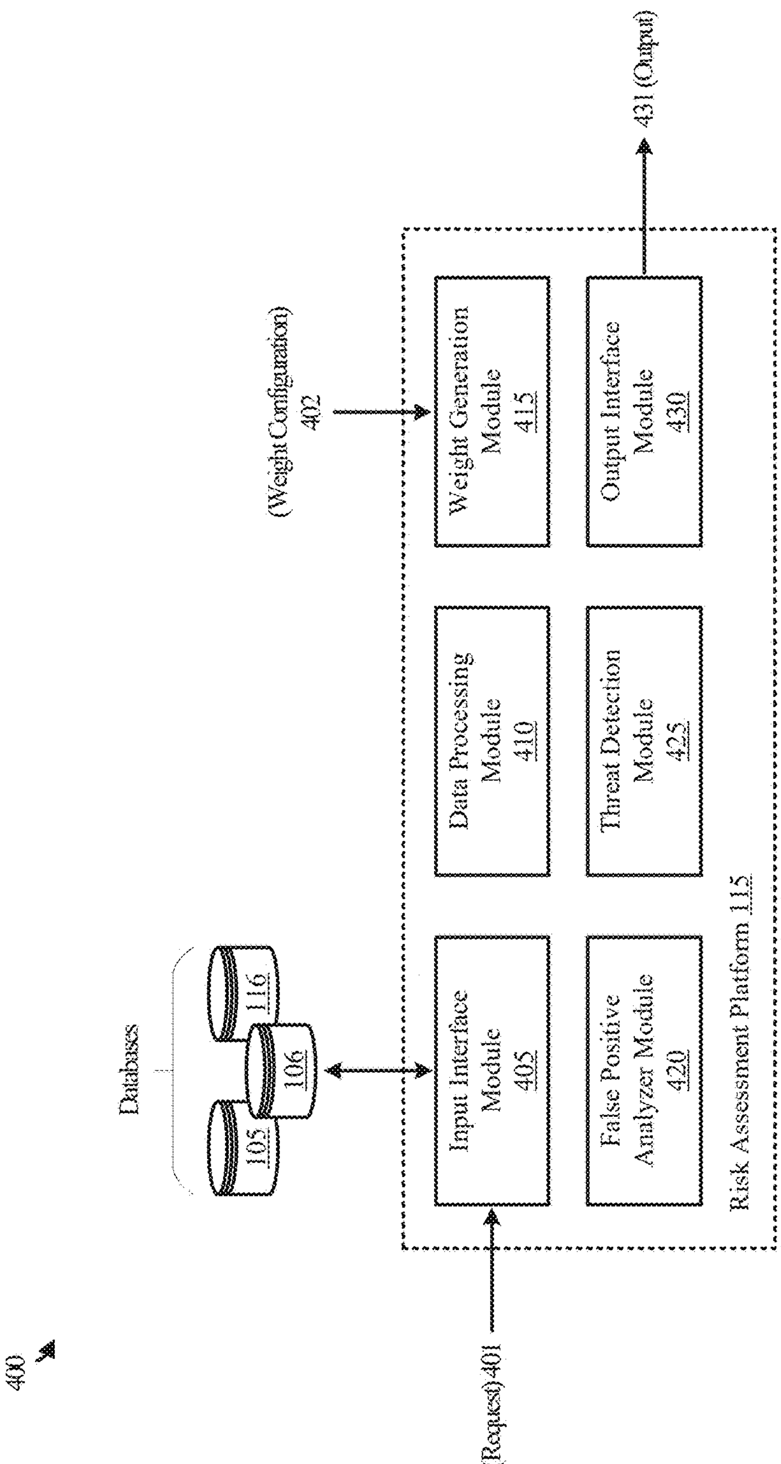

400
(Request) 401
Databases
105
106
116
Input Interface Module 405
False Positive Analyzer Module 420
Data Processing Module 410
Threat Detection Module 425
(Weight Configuration) 402
Weight Generation Module 415
Output Interface Module 430
Risk Assessment Platform 115
431 (Output)

FIGURE 4

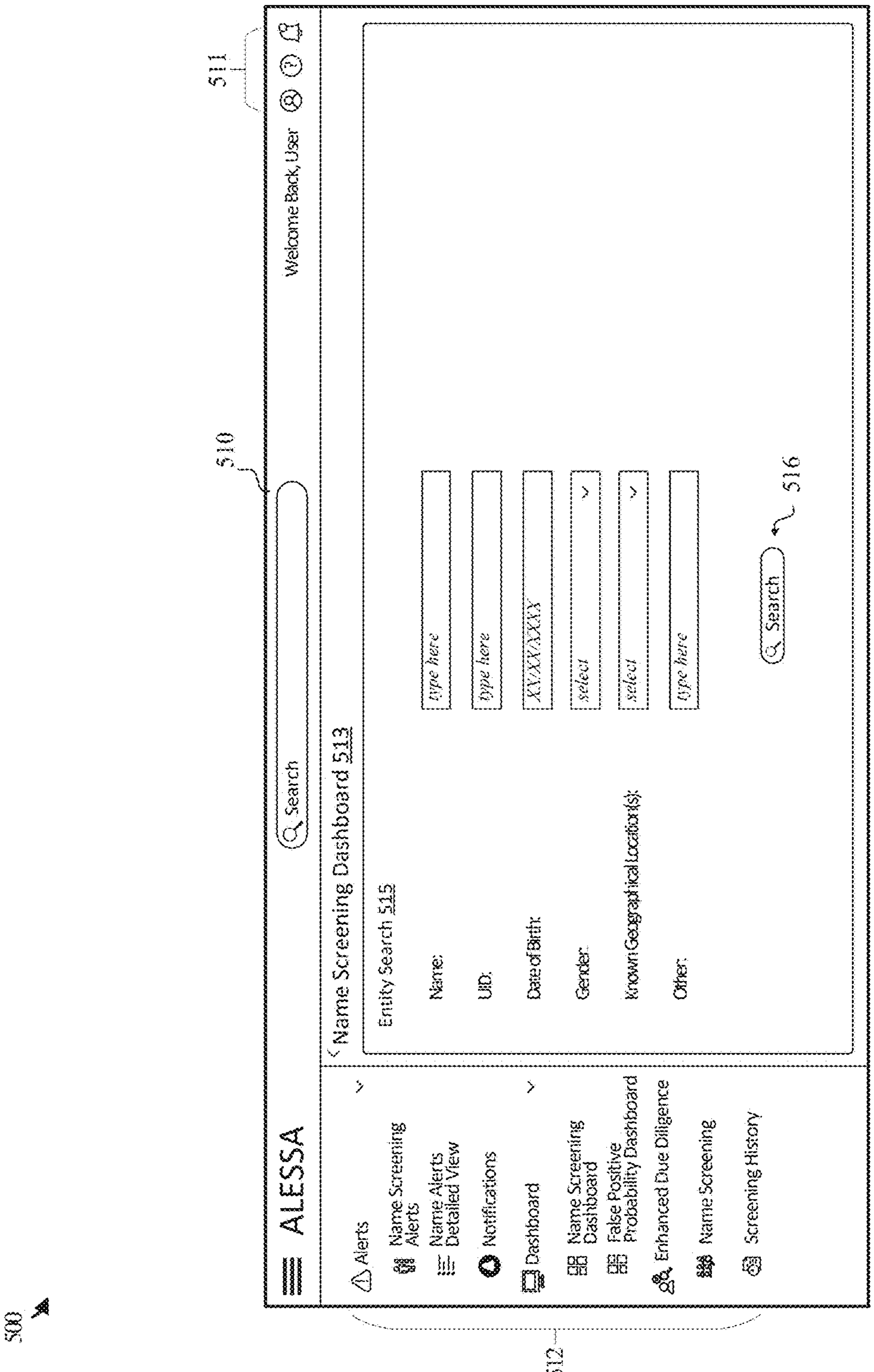

500
510
511
512
ALESSA
Search
Welcome Back, User
Alerts
Name Screening Alerts
Name Alerts Detailed View
Notifications
Dashboard
Name Screening Dashboard
False Positive Probability Dashboard
Enhanced Due Diligence
Name Screening
Screening History
Name Screening Dashboard 513
Entity Search 515
Name:
UID:
Date of Birth:
Gender:
Known Geographical Location(s):
Other:
type here
type here
XX/XX/XXXX
select
select
type here
Search
516

FIGURE 5A

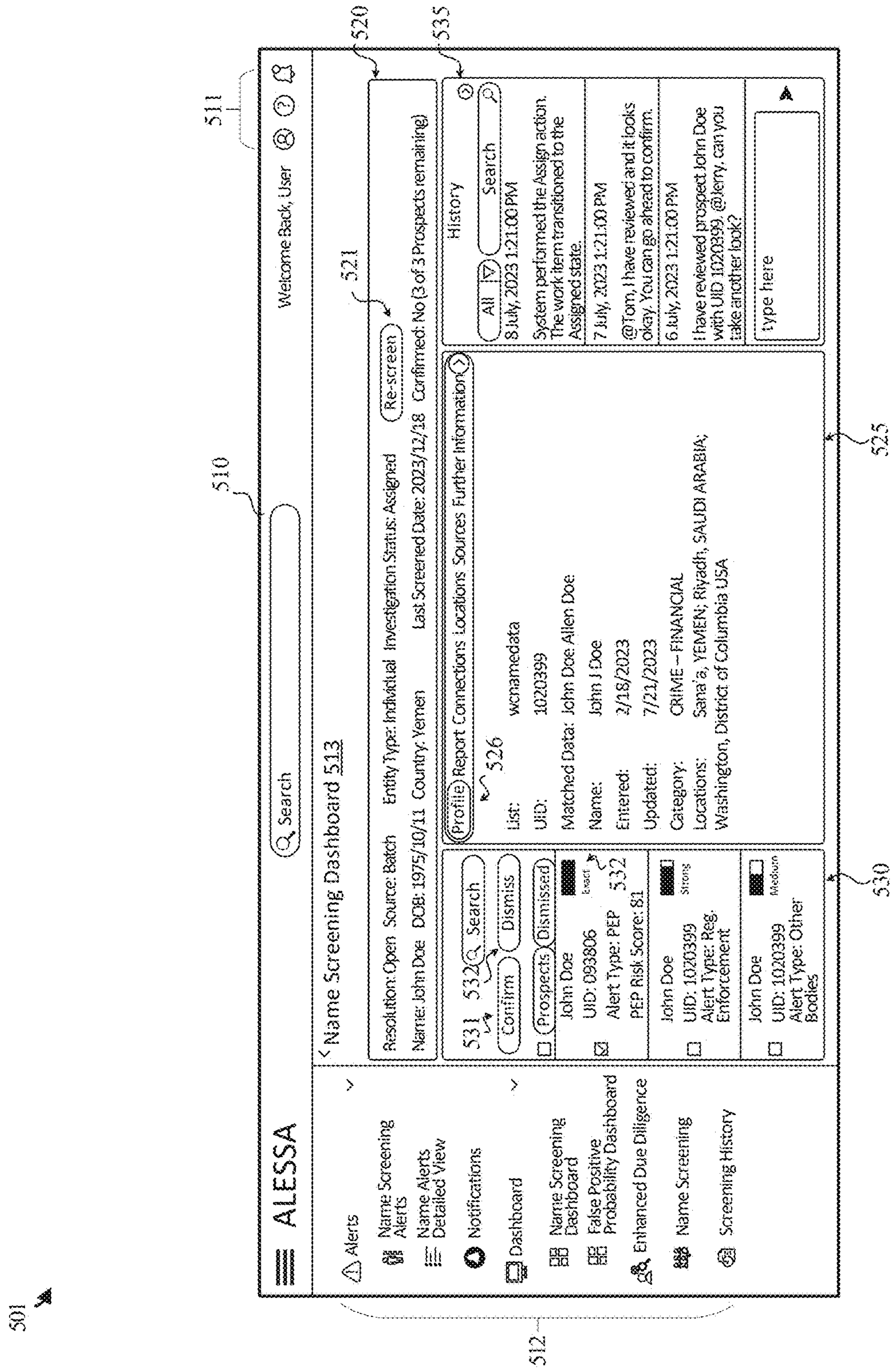

501
510
511
512
520
521
525
526
530
531
532
535
ALESSA
Search
Welcome Back, User
Alerts
Name Screening Alerts
Name Alerts Detailed View
Notifications
Dashboard
Name Screening Dashboard
False Positive Probability Dashboard
Enhanced Due Diligence
Name Screening
Screening History
Name Screening Dashboard 513
Resolution: Open Source: Batch Entity Type: Individual Investigation Status: Assigned
Name: John Doe DOB: 1975/10/11 Country: Yemen Last Screened Date: 2023/12/18 Confirmed: No (3 of 3 Prospects remaining)
Re-screen
Confirm
Search
Dismiss
Prospects
Dismissed
John Doe
UID: 093806
Alert Type: PEP
PEP Risk Score: 81
Exact
John Doe
UID: 1020399
Alert Type: Reg. Enforcement
Strong
John Doe
UID: 1020399
Alert Type: Other Bodies
Medium
Profile Report Connections Locations Sources Further Information
List: wcnamedata
UID: 1020399
Matched Data: John Doe Allen Doe
Name: John J Doe
Entered: 2/18/2023
Updated: 7/21/2023
Category: CRIME – FINANCIAL
Locations: Sana'a, YEMEN; Riyadh, SAUDI ARABIA; Washington, District of Columbia USA
History
All
Search
8 July, 2023 1:21:00 PM
System performed the Assign action. The work item transitioned to the Assigned state.
7 July, 2023 1:21:00 PM
@Tom, I have reviewed and it looks okay. You can go ahead to confirm.
6 July, 2023 1:21:00 PM
I have reviewed prospect John Doe with UID 1020399, @Jerry, can you take another look?
type here

FIGURE 5B

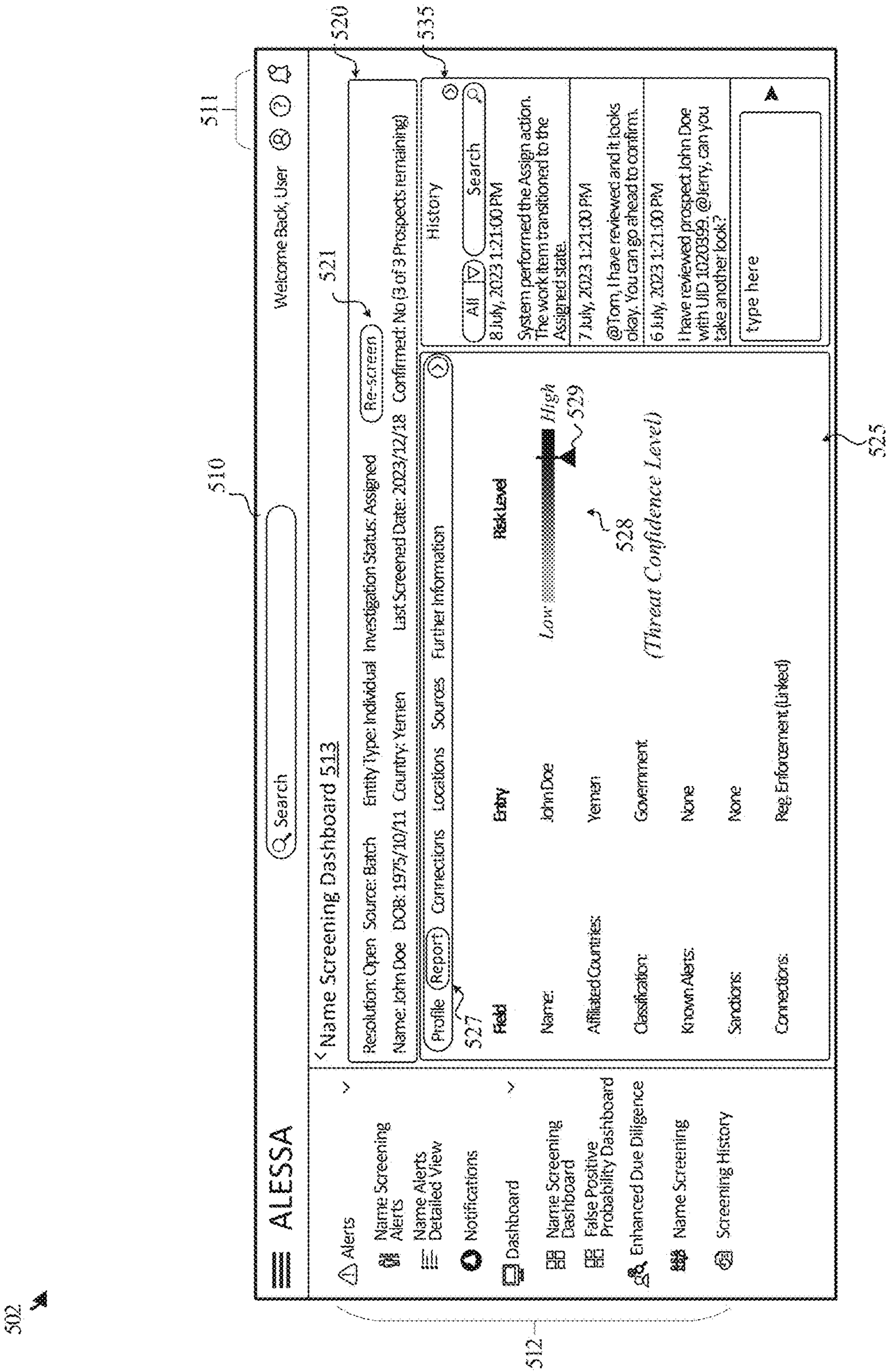

502
510
511
512
520
521
525
527
528
529
535
ALESSA
Search
Welcome Back, User
Alerts
Name Screening Alerts
Name Alerts Detailed View
Notifications
Dashboard
Name Screening Dashboard
False Positive Probability Dashboard
Enhanced Due Diligence
Name Screening
Screening History
Name Screening Dashboard 513
Resolution: Open Source: Batch Entity Type: Individual Investigation Status: Assigned
Name: John Doe DOB: 1975/10/11 Country: Yemen Last Screened Date: 2023/12/18 Confirmed: No (3 of 3 Prospects remaining)
Re-screen
Profile
Report
Connections
Locations
Sources
Further Information
Field
Entry
Risk Level
Name:
John Doe
Affiliated Countries:
Yemen
Classification:
Government
Known Alerts:
None
Sanctions:
None
Connections:
Reg. Enforcement (Linked)
Low
High
(Threat Confidence Level)
History
All
Search
8 July, 2023 1:21:00 PM
System performed the Assign action. The work item transitioned to the Assigned state.
7 July, 2023 1:21:00 PM
@Tom, I have reviewed and it looks okay. You can go ahead to confirm.
6 July, 2023 1:21:00 PM
I have reviewed prospect John Doe with UID 1020399. @Jerry, can you take another look?
type here

FIGURE 5C

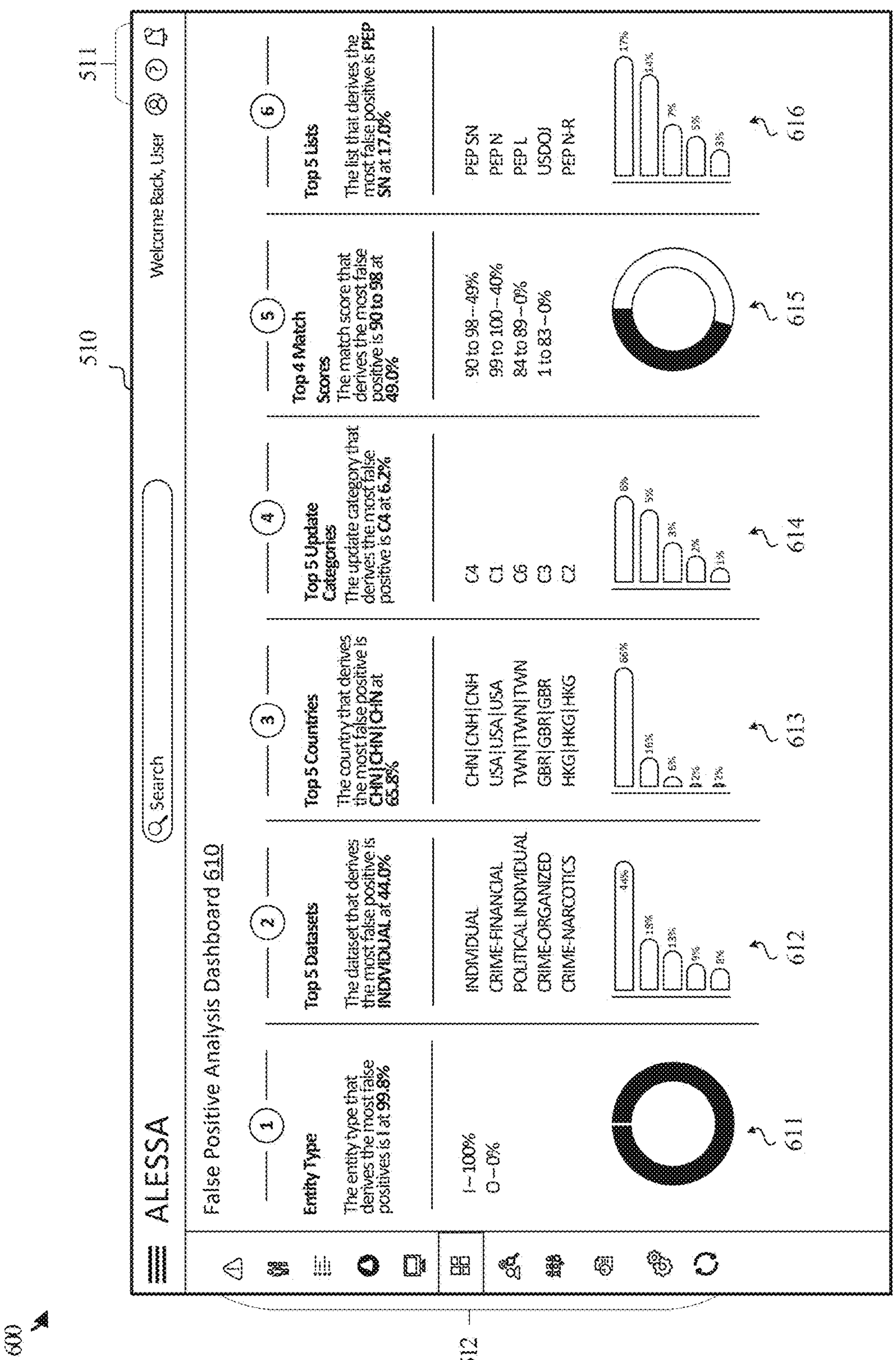

600
510
511
512
ALESSA
Search
Welcome Back, User
False Positive Analysis Dashboard 610
1
Entity Type
The entity type that derives the most false positives is I at 99.8%
I~100%
O~0%
611
2
Top 5 Datasets
The dataset that derives the most false positive is INDIVIDUAL at 44.0%
INDIVIDUAL
CRIME-FINANCIAL
POLITICAL INDIVIDUAL
CRIME-ORGANIZED
CRIME-NARCOTICS
44%
18%
13%
9%
8%
612
3
Top 5 Countries
The country that derives the most false positive is CHN|CHN|CHN at 65.8%
CHN|CNH|CNH
USA|USA|USA
TWN|TWN|TWN
GBR|GBR|GBR
HKG|HKG|HKG
66%
16%
6%
2%
2%
613
4
Top 5 Update Categories
The update category that derives the most false positive is C4 at 6.2%
C4
C1
C6
C3
C2
6%
5%
3%
2%
1%
614
5
Top 4 Match Scores
The match score that derives the most false positive is 90 to 98 at 49.0%
90 to 98~49%
99 to 100~40%
84 to 89~0%
1 to 83~0%
615
6
Top 5 Lists
The list that derives the most false positive is PEP SN at 17.0%
PEP SN
PEP N
PEP L
USDOJ
PEP N-R
17%
14%
7%
5%
3%
616

FIGURE 6

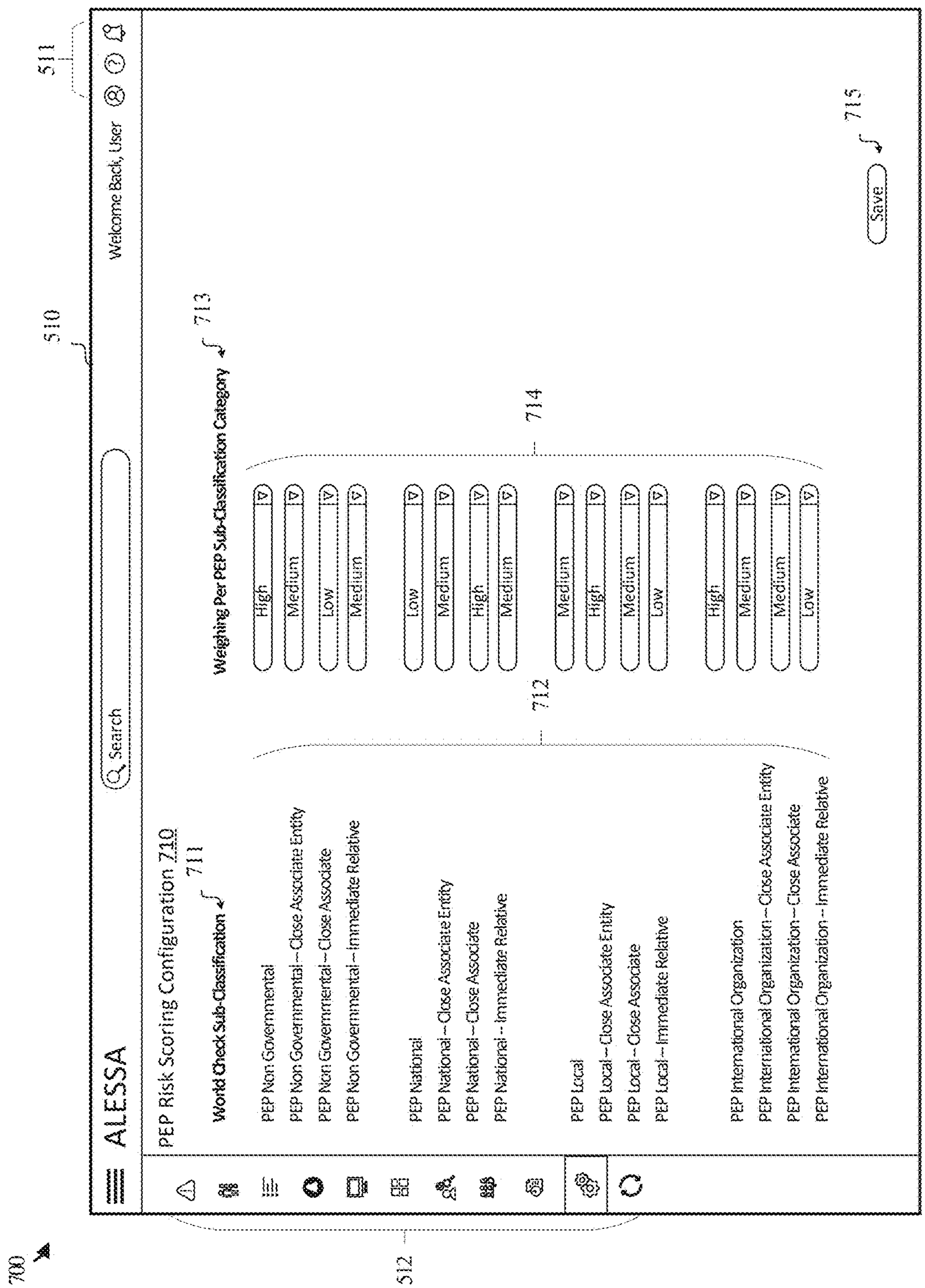

700
510
511
512
ALESSA
Search
Welcome Back, User
PEP Risk Scoring Configuration 710
711
World Check Sub-Classification
712
Weighing Per PEP Sub-Classification Category
713
714
PEP Non Governmental
PEP Non Governmental – Close Associate Entity
PEP Non Governmental – Close Associate
PEP Non Governmental – Immediate Relative
PEP National
PEP National – Close Associate Entity
PEP National – Close Associate
PEP National – Immediate Relative
PEP Local
PEP Local – Close Associate Entity
PEP Local – Close Associate
PEP Local – Immediate Relative
PEP International Organization
PEP International Organization – Close Associate Entity
PEP International Organization – Close Associate
PEP International Organization – Immediate Relative
High
Medium
Low
Medium
Low
Medium
High
Medium
Medium
High
Medium
Low
High
Medium
Medium
Low
Save
715

FIGURE 7

801

Range 810	Score 811
Low	0
Medium	55
High	76

802

Factor Number 815	Factor 816
1	Customer & Entity
2	Geographic Location

803

Identifier 820	Record Categories 821	Weight 822	Override Factor 823
1	Country CPI	40	
2	WC PEP Classification	30	
3a	Law Enforcement Alert	30	
3b	Regulatory Enforcement Alert	30	
3c	Other Bodies Alert	30	
3d	Sanctions Alert	76	Yes
4a	Law Enforcement Alert (Linked Entity)	20	
4b	Regulatory Enforcement Alert (Linked Entity)	20	
4c	Sanctions Alert (Linked Entity)	46	
4d	PEP Classification (Linked Entity)	20	
5	High Risk Countries	20	

FIGURE 8

901

Identifier	820	1
Factor Number	815	2
Record Category	821	Country CPI Rank
Override Factor	823	No
Weight	822	40
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Direct

902

Country Name 920	CPI Score 921	Rank 922	Normalized Score 923
Syria	13	178	99.0
South Sudan	13	178	99.0
Somalia	12	180	99.0
Venezuela	14	177	98.0
Yemen	16	176	97.0
...	...	...	...
Norway	84	4	2.0
Singapore	83	5	2.0
Finland	87	2	1.0
New Zealand	87	2	1.0
Denmark	90	1	0.0

FIGURE 9

1001

Identifier	820	2
Factor Number	815	2
Record Category	821	PEP Classification
Override Factor	823	No
Weight	822	30
Delay Duration of Risk	915	
Duration of Risk	916	
Module	917	Match

FIGURE 10A

1002

Condition Threshold 1010	Score 1011
PEP Non-Government (NG) PEP NG-AE State Owned Enterprise PEP N PEP N-AE Sub-National Government PEP N-A PEP N-R PEP SN PEP N-AE Sub-National PEP SN-A	100
PEP-L PEP L-AE Non-Governmental PEP L-A PEP L-R PEP IO PEP IO-AE Regional Organization PEP IO-A PEP RO PEP RO-AE National PEP RO-A	67
SOE SIE IOS	33
PEP SN-R PEP NG-R PEP IO-R PEP RO-R	0

FIGURE 10B

1101

Identifier	820	3a
Factor Number	815	2
Record Category	821	Law Enforcement
Override Factor	823	No
Weight	822	20
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

1102

Identifier	820	3b
Factor Number	815	2
Record Category	821	Reg. Enforcement
Override Factor	823	No
Weight	822	20
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

FIGURE 11A

1103

Identifier	820	3c
Factor Number	815	2
Record Category	821	Other Bodies
Override Factor	823	No
Weight	822	20
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

1104

Identifier	820	3d
Factor Number	815	2
Record Category	821	Sanctions
Override Factor	823	Yes
Weight	822	76
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

FIGURE 11B

1201

Identifier	820	4a
Factor Number	815	2
Record Category	821	Law Enforcement (Linked)
Override Factor	823	No
Weight	822	20
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

1202

Identifier	820	4b
Factor Number	815	2
Record Category	821	Reg. Enforcement (Linked)
Override Factor	823	No
Weight	822	20
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

FIGURE 12A

1203

Identifier	820	4c
Factor Number	815	2
Record Category	821	Sanctions
Override Factor	823	No
Weight	822	46
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

1204

Identifier	820	4d
Factor Number	815	2
Record Category	821	PEP
Override Factor	823	No
Weight	822	20
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

FIGURE 12B

1301

Identifier	820	5
Factor Number	815	2
Record Category	821	High Risk Countries
Override Factor	823	No
Weight	822	20
Delay Duration of Risk	915	0
Duration of Risk	916	0
Module	917	Match

1302

Condition Threshold 1310	Score 1311
Russian Federation	100
China	50

FIGURE 13

1401

Factor 1410		Factor Score 1411	Weight Score 1412
Countries 1413	Iraq – 86% Jordan – 55% Syria – 99% UAE – 29%	99%	39.6
Category 1414	PEP N	100%	30
Alerts 1415	Sanctions	100%	76
	Reg. Enforcement	100%	30
	Law Enforcement	100%	30
	Other Bodies	100%	30
Linked Entities 1416	PEP	100%	76
	Sanctions	100%	30
	Reg. Enforcement	100%	30
	Law Enforcement	100%	30
High Risk Countries 1417	Syria	100%	20
Total Score 1418 (excluding override factors)			285.6

FIGURE 14A

1102

Factor 1410		Factor Score 1411	Weight Score 1412
Countries 1413	Indonesia -- 72%	72%	28.8
Category 1414	PEP L-R	67%	20
Alerts 1415	None	0%	0
Linked Entities 1416	PEP	100%	20
High Risk Countries 1417	None	0%	0
Total Score 1418 (excluding override factors)			68.8

FIGURE 14B

1403

Factor 1410		Factor Score 1411	Weight Score 1412
Countries 1413	Denmark -- 0%	0%	0
Category 1414	PEP N-AE	0%	0
Alerts 1415	None	0%	0
Linked Entities 1416	PEP	100%	20
High Risk Countries 1417	None	0%	0
Total Score 1418 (excluding override factors)			20

FIGURE 14C

1404

Factor 1410		Factor Score 1411	Weight Score 1412
Countries 1413	Ghana – 60%	60%	24
Category 1414	PEP L	67%	20
Alerts 1415	None	0%	0
Linked Entities 1416	None	0%	0
High Risk Countries 1417	None	0%	0
Total Score 1418 (excluding override factors)			44
Deceased Override 1419			30

FIGURE 14D

500

Geography 1510	Points 1514
Most Popular Countries	20
Popular Countries	10
Other Countries	5

1501

Last Name Commoness 1511	Points 1514
Top 100 Caucasian Names List	40
Top 50 Asian Names List	40
Top 50 Hispanic Names List	40
Top 30 Indian Names List	40

1502

Missing Fields 1512	Points 1514
Date of Birth	25
Gender	25

1515

Score	Result
86+	Very High
70-85	High
0-69	Low

FIGURE 15

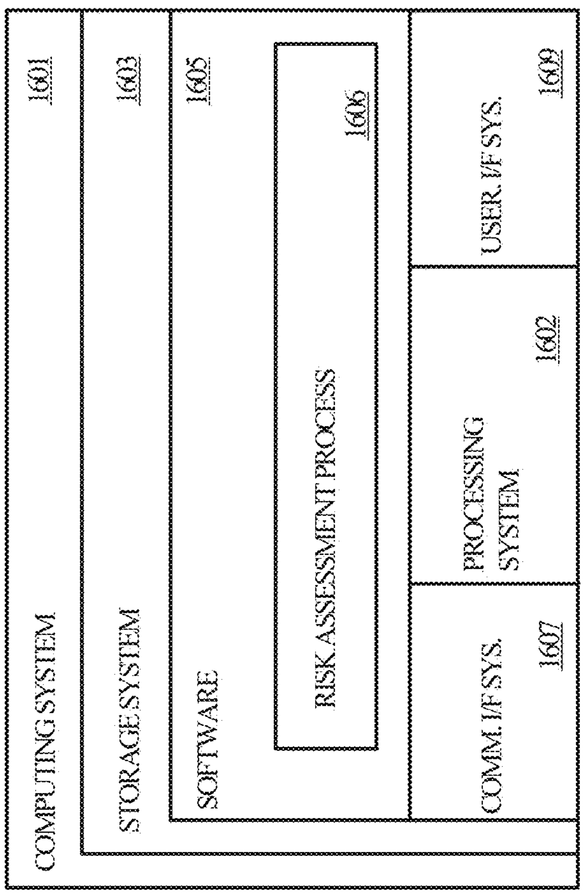

COMPUTING SYSTEM
1601
STORAGE SYSTEM
1603
SOFTWARE
1605
RISK ASSESSMENT PROCESS
1606
COMM. I/F SYS.
1607
PROCESSING SYSTEM
1602
USER. I/F SYS.
1609

FIGURE 16

SYSTEMS, METHODS, AND SOFTWARE FOR ENHANCED RISK ASSESSMENT

RELATED APPLICATIONS

[0001] This application claims the priority benefit of U.S. Application No. 63/562,958, filed Mar. 8, 2024, entitled "SCORING METHODOLOGY FOR POLITICALLY-EXPOSED-PERSONS," which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This relates generally to risk assessment and analysis, and more particularly, to risk analysis computation hardware and software.

BACKGROUND

[0003] Financial and non-financial institutions are required to screen various individuals or entities to minimize risk to the institutions and prevent financial crimes, such as money laundering, bribery, corruption, and terrorist financing. Screening helps institutions comply with global regulations and avoid legal, financial, and reputational risks. Failure to screen or proceeding to transact with high-risk individuals may result in penalties, loss of credibility, and even legal action.

[0004] Institutions often categorize individuals and entities by risk levels to adjust screening and compliance efforts accordingly. However, few solutions exist to accurately detect and categorize risk levels. Problematically, this may lead institutions to miss high-risk individuals and entities during screening exercises or unnecessarily screen lower-risk individuals and entities.

[0005] Despite filtering efforts to detect and review high-risk individuals and entities subject to higher review scrutiny, the amount of data to be screened is vast, often duplicative, and stored across several databases. As such, screening requires significant manual effort to obtain necessary records from the various sources, compile the records into consolidated profiles, and review the profiles for potential risks. Once the data is compiled and categorized, there may be numerous records that share the same name, geographical location, and roles, thus creating additional work to sift through false positive matches.

SUMMARY

[0006] Disclosed herein are improvements to risk assessment, false positive detection, and risk analysis visualization of entities with respect to commercial dealings therewith. In an example embodiment, a risk assessment platform is provided that includes one or more computer-readable storage media and program instructions stored on the one or more computer-readable storage media that, based on being read and executed by a processing device, direct the processing device to perform various functions. Specifically, the program instructions direct the processing device to receive an input with an indication of a target entity subject to a risk assessment, identify, based on the input, multiple possible instances of the target entity from a database, and for each possible instance of the multiple possible instances, evaluate a strength of a match of the possible instance to the target entity for display thereof on a graphical user interface. The program instructions may also direct the processing device to display various visual representations on a graphical user interface, such as a visual representation of the multiple possible instances, corresponding strengths of matches, and threat confidence levels, among other information.

[0007] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. It may be understood that this Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** illustrates an example operating environment for performing risk assessment and analysis in accordance with some embodiments of the present technology.

[0009] FIG. **2**A illustrates a series of steps for performing and displaying a risk assessment in accordance with some embodiments of the present technology.

[0010] FIG. **2**B illustrates a series of steps for performing and displaying a false positive probability analysis in accordance with some embodiments of the present technology.

[0011] FIG. **3** illustrates an example sequence diagram demonstrating access and data flow between elements of a system in accordance with some embodiments of the present technology.

[0012] FIG. **4** illustrates an example block diagram of a risk assessment platform in accordance with some embodiments of the present technology.

[0013] FIGS. **5**A, **5**B, and **5**C illustrate aspects of an example user interface associated with a risk assessment platform in accordance with some embodiments of the present technology.

[0014] FIG. **6** illustrates an aspect of an example user interface associated with a risk assessment platform in accordance with some embodiments of the present technology.

[0015] FIG. **7** illustrates an aspect of an example user interface associated with a risk assessment platform in accordance with some embodiments of the present technology.

[0016] FIGS. **8**, **9**, **10**A, **10**B, **11**A, **11**B, **12**A, **12**B, **13**, **14**A, **14**B, **14**C, **14**D, and **15** illustrate example tables including configurable information used in risk analyses in accordance with some embodiments of the present technology.

[0017] FIG. **16** illustrates a computing device that may be used in accordance with some examples of the present technology.

[0018] The drawings are not necessarily drawn to scale. In the drawings, like reference numerals designate corresponding parts throughout the several views. In some examples, components or operations may be separated into different blocks or may be combined into a single block.

DETAILED DESCRIPTION

[0019] Technology is disclosed herein that resolves the above issues with enhanced systems, methods, and software for risk assessment and analysis, threat detection, and false positive detection with respect to entities and financial dealings therewith. Risk-based screening techniques described herein relate to screening individuals and entities with a potential propensity to commit financial crimes, such as bribery, money laundering, and the like, screening transactions between individuals, entities, and institutions, and screening individuals and entities against alerts lists, like sanctions lists (e.g., OFAC, EU Sanctions, UN Sanctions), among other types of screening and risk analysis. Such screening processes help detect suspicious activities and high-risk transactions, prevent corruption and bribery, and protect financial institutions, non-financial institutions, and businesses from legal penalties, financial losses, and reputational damages.

[0020] In existing solutions, screening of individuals and entities, for example, is performed manually by obtaining various information about the individuals and entities, checking alerts and sanctions lists for notices associated with the individuals and entities, and identifying other individuals and entities connected to the individuals and entities subject to screening. The data is often spread across several databases making generating records for each individual or entity tedious. Some databases include the same or similar information as other databases, which may lead to the creation of duplicative or overlapping records for the individuals and entities, and ultimately, false positive records. Once an institution compiles a record(s) for an individual or entity, a user manually categorizes the record by a risk level.

[0021] High-risk records often requires enhanced due diligence meaning a user must perform extensive investigations into an individual's or entity's financial history, transaction patterns, connections, and source of funds.

[0022] To alleviate manual effort, reduce false positive records, and enhance the user experience with respect to screening processes, a risk assessment platform is described herein that accesses various databases, standardizes records and associated values obtained from the various databases, creates profiles for the individuals and entities subject to screening, and visualizes various aspects of the profiles, such as a threat confidence level and a false positive probability level. The threat confidence level associated with a profile may indicate a potential risk for suspicious or unlawful activity based on the individual's or entity's information, such as location, occupation, role, connections, and the like. The false positive probability level associated with a profile visually indicates to a user a strength of a match of a particular record relative to a queried name or identifier.

[0023] In an example embodiment, a risk assessment platform is provided that includes one or more computer-readable storage media and program instructions stored on the one or more computer-readable storage media that, based on being read and executed by a processing device, direct the processing device to perform various functions. Specifically, the program instructions direct the processing device to receive an input with an indication of a target entity subject to a risk assessment, identify, based on the input, multiple possible instances of the target entity from a database including a plurality of entities, and for each possible instance of the multiple possible instances, evaluate a strength of a match of the possible instance relative to the target entity based on applying a set of criteria against the indication of the target entity and comparing results of the application of the set of criteria with an indication of the possible instance. The program instructions may also direct the processing device to display, on a graphical user interface, a visual representation of the multiple possible instances and corresponding strengths of matches. Additionally, the program instructions may direct the processing device to, in response to a selection of one or more of the multiple possible instances, add the one or more of the multiple possible instances to a queue associated with a further risk assessment.

[0024] In another example embodiment, a risk assessment platform is provided that includes one or more computer-readable storage media and program instructions stored on the one or more computer-readable storage media that, based on being read and executed by a processing device, direct the processing device to perform various functions. Specifically, the program instructions direct the processing device to receive an input with an indication of a target entity subject to a risk assessment, and identify, based on the input, multiple possible instances of the target entity from a database including a plurality of entities. The program instructions also direct the processing device to display, on a graphical user interface, a visual representation of the multiple possible instances, corresponding strengths of matches relative to the target entity, and corresponding risk profiles. The program instructions also direct the processing device to, in response to a selection of a risk profile of one of the multiple possible instances, display, on the graphical user interface, a visual representation of a risk assessment generated for the risk profile based on performing a risk analysis on data associated with the risk profile accessed through one or more databases, wherein the visual representation of the risk assessment includes an indication of a threat confidence level associated with the risk profile.

[0025] In yet another example embodiment, a risk assessment system is provided that includes a memory, and a processor coupled with executable instructions forming modules of the risk assessment system and configured to execute the modules of the risk assessment system to produce a risk assessment. The modules executed by the processor include an input interface configured to obtain, from one or more databases, risk and governance data associated with an entity subject to the risk assessment, and obtain, from a user device, an input comprising an indication of a target entity, an output interface configured to output the risk assessment of the target entity, a data processing module configured to receive the risk and governance data associated with the target entity from the input interface and identify a subset of records having associated values, convert the values associated with the subset of records of the risk and governance data from a first format to a second format, and provide the converted values to a weight generation module, the weight generation module configured to receive the risk and governance data and the converted values from the data processing module, generate a set of weights for the risk and governance data, wherein the set of weights comprises a first subset of weights including the converted values, and a second subset of weights, and provide the set of weights to a scoring module, and the scoring module configured to generate the risk assessment of the target entity based on applying the set of weights to the risk and governance data and provide the risk assessment to the output interface.

[0026] Various embodiments of the present technology provide for a wide range of technical effects, advantages, and/or improvements to computing systems and components.

[0027] For example, various embodiments may include one or more of the following technical effects, advantages, and/or improvements: 1) entity record aggregation and standardization; 2) entity record indexing efficiency; 3) false positive detection accuracy and efficiency; 4) risk/threat level categorization accuracy and efficiency; 5) user experience and user interface enhancement with respect to at least false positive detection and risk/threat level identification; and 6) confirmation, dismissal, and enqueuing of entity records for further screening and risk assessment (e.g., enhanced due diligence) via automation, machine learning, and/or Artificial Intelligence.

[0028] In particular, the advantages of the technology disclosed herein include systems, methods, software, and devices for performing risk assessment and analysis based on personal, geographical, occupational, and financial data obtained from various sources. For institutions subject to screening regulations, the proposed solution can reduce the number of false positive records flagged for review, increase the accuracy with which records are categorized as requiring review, and improve the user experience related to screening, risk assessment, and risk analysis via automation and visualization.

[0029] While various embodiments of the present technology relate to screening individuals or entities, such as politically-exposed persons (PEPs), with respect to financial and commercial dealings, it may be appreciated that the systems, methods, software, and devices described herein may be utilized in other applications, such as in other types of financial screening applications, transaction screening applications, insurance screening applications, medical screening applications, and the like.

[0030] Moving now to the Figures, FIG. **1** illustrates an example operating environment for performing risk assessments and analyses in accordance with some embodiments of the present technology. FIG. **1** shows operating environment **100**, which includes database **105**, database **106**, Internet **110**, risk assessment platform **115**, database **116**, and user devices **120** and **125**.

[0031] In various embodiments, risk assessment platform **115** is representative of a system, device, application, server, or the like capable of processing data from various databases, performing risk assessments and analyses, and outputting indications (e.g., visual, graphical) associated with risk assessments and analyses for entities (e.g., individuals, organizations) as requested by users. Risk assessment platform **115** may be configured to perform risk assessment and analysis operations, such as those of methods **201** and **202** of FIGS. **2**A and **2**B, respectively. As such, risk assessment platform **115** may be implemented in the context of software, hardware, and/or firmware, as well as combinations and variations thereof.

[0032] To perform risk assessments and analyses of entities, risk assessment platform **115** accesses databases **105** and **106** via respective interfaces and over a communication network (i.e., via Internet **110**), and obtains data associated with entities from databases **105** and **106**. Examples of the interfaces with which risk assessment platform **115** accesses databases **105** and **106** includes application programming interfaces (APIs), RESTful APIs, file transfer protocols (FTPs), and the like.

[0033] In various embodiments, database **105** is representative of a repository, data storage service, or another type of memory device that stores a first set of data associated with entities, such as corruption perception index (CPI) data. For example, database **105** may be a database hosted and provided by Transparency International, which may be accessible via Internet **110**. The CPI data stored in database **105** may include data aggregated from a number of different sources that provide perceptions of businesspeople and country experts of the level of corruption in the public sector of various countries. In various embodiments, the CPI data may include data fulfilling criteria to quality as valid data. The criteria may include several factors related to whether the data source quantifies risks or perceptions of corruption in the public sector, bases data aggregation on a reliable and valid methodology, is a reputable organization, allows for sufficient variation of scores to distinguish between countries, ranks a substantial number of countries, considers only the assessments of country experts or businesspeople, and is regularly updated.

[0034] In various embodiments, the CPI data may include a list, table, or other data structure including country names, CPI scores associated with the countries, and ranks of the countries relative to one another. The CPI score may correspond to risk level. In database **105**, the CPI scores may be provided on a 100% scale with corruption perception decreasing as the score increases. For example, in a previous report (**2022** CPI Report), Denmark may have the highest CPI score with a score of 90, and thus, the lowest rank of 1 among all countries. Risk assessment platform **115** may obtain the CPI data and perform scoring standardization operations on the CPI data for use in scoring processes. More specifically, risk assessment platform **115** may normalize the CPI scores such that the highest score is set to a value of 0 and the lowest score is set to 100. Risk assessment platform **115** may use the following equation to normalize the CPI scores:

$$\text{Normalized } CPI \text{ Score} = \left(1 - \frac{CPI \text{ Score} - \text{Minimum } CPI \text{ Score}}{\text{Maximum } CPI \text{ Score} - \text{Minimum } CPI \text{ Score}}\right) \times 100$$

[0035] Examples of the CPI data, including CPI scores, country names, country ranks, and normalized CPI scores is shown and described in table **902** of FIG. **9**.

[0036] Risk assessment platform **115** may further perform mapping processes to map CPI country codes to other naming conventions. For example, database **105** may include country names using a ISO3 code, and risk assessment platform **115** may obtain such codes from respective data fields, extract the codes from the data fields, map the codes to different fields with a different naming convention, and store the country names in database **116**.

[0037] Database **106** may be representative of another database, data storage service, or another type of memory that stores data related to individuals or entities. For example, such data may include politically-exposed persons (PEP) data, financial transaction data, personal information, and connected entity data, among other information. In various embodiments, database **106** is a database hosted and provided by World-Check, Dow Jones, or Rzolut. The data stored in database **106** may include a number of individuals or entities, a unique identifier associated with each individual or entity, and a number of records associated with each individual or entity. Examples of the records include a date of birth (applicable for individuals), a gender (applicable for individuals), locations (e.g., place of birth, location of employment, location of residence, location of domicile), countries corresponding to nationality and/or citizenship, a classification or sub-classification, alert types, and the like. More specifically, the classifications may include various categories, or abbreviations thereof, such as non-governmental, national, subnational government, local, international organization, regional organization, state-owned organization, state-invested organization, and instrumentality of state. Some of the classifications may further include sub-classifications, such as close associate entity, close associate, and immediate relative. The alert types may include a number of categories, such as a law enforcement alert, a regulatory enforcement alert, a sanctions alert, a high-risk country alert, and other alerts. Each of the alert types may further include a sub-type, such as a linked entity alert.

[0038] In various embodiments, risk assessment platform **115** may be configured to obtain the data from database **106** and aggregate this data with the data obtained from database **105** to create an indexed data set. More particularly, risk assessment platform **115** can use the aggregated data to generate a profile for each individual and/entity identified including various personal, geographical, occupational, and other data.

[0039] To extract various records from the data obtained in database **106**, risk assessment platform **115** may be configured to extract information from fields of the PEP data using one or more extraction functions. For example, risk assessment platform **115** may use the following extraction formula to obtain various records, such as place of birth:

$$extractall('\backslash\backslash s*([':\$]+)(?:;|\$)')$$

[0040] Other variations and combinations of extraction formulas may be used to obtain and aggregate records of the PEP data.

[0041] In some embodiments, database **106** may also include common name data that includes a list of common surnames for each of multiple ethnic groups. In some embodiments, the common name data may be obtained from a different database via a different interface. The common name data may include a list of the top 100 most common traditionally white or Caucasian last names, a list of the top 50 most common Asian last names, a list of the top 50 most common Hispanic last names, and a list of the top 30 most common Indian last names, among other lists. Upon aggregation of the common name data with the PEP data, risk assessment platform **115** may include an indication of whether the name of an individual matches a common last name found in one or more of the lists of the common name data. Risk assessment platform **115** can store the aggregated data in database **116**, representative of a local or cloud database accessible by risk assessment platform **115** different from database **105** and database **106**.

[0042] For each record in the aggregated data, such as the individual or entity name, the date of birth, the gender, the cities, states, countries, classifications, alert types, and the like, risk assessment platform **115** may be configured to determine a weight and a set of conditions or criteria against which to apply the records. A weight may include a value corresponding to a maximum score achievable if a record matches a condition. The set of conditions may include different threshold levels or direct matches. For example, risk assessment platform **115** may determine a weight for country records having a value of 40. If a risk profile includes the highest-risk country (e.g., based on the CPI data), risk assessment platform **115** may determine that the country record matches a high-risk country condition and award the maximum score for that weight (i.e., 40). If a risk profile includes a country that is not the highest-risk country, risk assessment platform **115** may determine that the country does not match the high-risk country condition, but instead matches a different risk condition, and may award a lesser score for that weight. If the condition is a direct match condition, risk assessment platform **115** may award 100% of the weight if the record includes a direct match or 0% of the weight if the record does not include a direct match.

[0043] Risk assessment platform **115** may determine the weights based on a level of corruption or risk associated with the record. For example, risk assessment platform **115** may assign a weight with a high value for a sanctions alert and a weight with lower value for a law enforcement alert relative to the value of the sanctions alert weight as sanctions may be considered to be more high-risk with respect to PEPs and associated potential corruption. Sample values for the weights assigned to each record may be shown and described in table **803** of FIG. **8**.

[0044] In some embodiments, risk assessment platform **115** may determine the weights based on a request from a user, such as request **121-1** from user device **120** (e.g., a computer, a tablet, a smart phone) or request **121-2** from user device **125** (e.g., a computer, a tablet, a smart phone). In such embodiments, a user of the user device may input custom weight values as part of the request, and risk assessment platform **115** may use the custom weight values for scoring operations and store the custom weights in database **116**.

[0045] Risk assessment platform **115** may determine the conditions based on a complexity of the record. For example, risk assessment platform **115** may use a multi-level condition with different threshold levels for a record including multiple sub-categories or sub-classifications (e.g., PEP classification). On the other hand, risk assessment platform **115** may use a direct matching condition (e.g., an all-or-nothing condition) for a record including few options (e.g., a high-risk country record).

[0046] After determining the weights and corresponding conditions for the records, risk assessment platform **115** may be configured to, for each profile, apply the conditions against the records found in the profile, determine a value for each weight based on applying the conditions against the records, and generate a risk-based score based on the total of the values. Risk assessment platform **115** can compare the risk-based score to threshold scores and provide an indication of the risk-based score to a user device (e.g., user device **120**) based on the risk-based score exceeding one or more of the threshold scores. The indication may include a status indicator, a pop-up alert, a message alert, or the like that can be displayed on a user interface of the user device.

[0047] Risk assessment platform **115** may also be configured to flag a record as an override factor. An override factor may correspond to the highest, or most important, risk record in the aggregated data, such that if a record flagged as an override factor exists in a profile, risk assessment platform **115** can award an amount of points that meets or exceeds a threshold score, and the profile can be indicated as high-risk despite the presence, or lack thereof, of other records in the profile.

[0048] By way of a first example, a user of user device **120** provides request **121-1** to risk assessment platform **115** for a risk assessment of one or more entities. Request **121-1** may include an indication of an entity, such as a name and/or a unique identifier (ID). Risk assessment platform **115** receives request **121-1**, identifies one or more instances associated with the name and/or unique ID, and obtains the instances from database **116**. Risk assessment platform **115** may identify the weights and conditions associated with the records of all the aggregated data, identify the records associated with the requested entity, and evaluate the identified records against the conditions to generate a threat confidence level associated with the requested profile based on the weights and conditions. Risk assessment platform **115** can then provide indication **122-1** to user device **120** for instantiation on a user interface thereof. Providing the risk-based score indication **122-1** to user device **120** may entail providing and/or displaying a notification or indication to user device **120**, or a peripheral thereof (e.g., a screen, a speaker). The notifications or indications of risk-based score indication **122-1** may differ based on whether the risk-based score falls below, meets, or exceeds threshold scores.

[0049] In some examples, request **121-1** may include a name of an entity without a unique ID, and risk assessment platform **115** may identify multiple similar profiles based on request **121-1**. In such examples, risk assessment platform **115** may obtain records for each of the similar profiles and evaluate the profiles against criteria to determine a strength of match for each profile relative to the request profile. In some embodiments, this may entail performing a comparison of records between a similar profile and the request profile to determine which is most similar (e.g., has the most matching records). In some embodiments, risk assessment platform **115** may assign weights to each record and apply the weights to identified records in the similar profiles as well as identified records in the request profile. Then, risk assessment platform **115** may compare the results of the application of the weights to determine a strength of match. Indication **122-1** may include an indication of the strength of match, such that a user of user device **120** may view visual or graphical indicators corresponding to probabilities of false positive profiles relative to the requested profile.

[0050] By way of a second example, a user of user device **125** provides request **121-2** to risk assessment platform **115** for a risk-based score of one or more entities. Request **121-2** may include a name of an entity, a unique identifier (ID) of the PEP, and a custom set of weights for each record, or record type, of the data. Risk assessment platform **115** may receive request **121-2**, identify one or more profiles associated with the name and/or unique ID and obtain the profile from database **116**. Risk assessment platform **115** may identify the weights and conditions associated with the records of all the aggregated data based on request **121-2**, identify the records associated with the requested profile, and evaluate the identified records against the conditions to generate a threat confidence level based on the custom weights and conditions. Risk assessment platform **115** can then provide indication **122-2** to user device **125**. The threat confidence level, and consequently indication **122-2**, may differ from the threat confidence level and indication **122-1**, respectively, based on differences between the sets of weights used by risk assessment platform **115**. Advantageously, risk assessment platform **115** can provide dynamic risk-based scores based on weights that may be changed based on criteria a user determines to be important, risky, or the like when screening entities and analyzing risks thereof.

[0051] In some embodiments, risk assessment platform **115** may be configured to obtain data from respective databases, aggregate the data, and apply a set of weights and conditions to the data on a regular, automated basis. In some embodiments, however, risk assessment platform **115** may be configured to perform such operations in response to receiving a request. Other variations or combinations may be contemplated.

[0052] FIGS. **2**A and **2**B illustrate methods **201**, and **202**, respectively, which reference elements of operating environment **100** of FIG. **1**. Method **201** of FIG. **2**A illustrates a series of steps for performing and displaying a risk assessment in accordance with some embodiments of the present technology. FIG. **2**B illustrates a series of steps for performing and displaying a false positive probability analysis in accordance with some embodiments of the present technology. Methods **201** and **202** may be implemented by risk assessment platform **115** of FIG. **1**. Accordingly, methods **201** and **202** may be implemented in hardware, firmware, software, or combinations or variations thereof.

[0053] Referring first to method **201** of FIG. **2**A, in operation **205**, risk assessment platform **115** receives an input (e.g., request **121-1**) from a user. The input includes an indication of a target entity subject to a risk assessment. More specifically, the indication may include a name of the target entity, an identifier associated with the target entity, a location associated with the target entity, a date of birth of the target entity, and other personal, occupational, and geographic information associated with the target entity. The user may provide the input to risk assessment platform **115** via a user interface of a user device (e.g., user device **120**) in communication with risk assessment platform **115** over a communication network.

[0054] In operation **210**, risk assessment platform **115** identifies possible instances of the target entity specified in the input. To do so, risk assessment platform **115** may first access one or more databases (e.g., database **105**, database **106**, database **116**) and query the database(s) for records matching the indication of the target entity. For example, risk assessment platform **115** performs a query for the target entity using the name, identifier, and location of the target entity. The query results may include multiple entities that share the same or similar names, locations, and other identifying information. Some of these entities may be false positives, while others may correspond to the target entity.

[0055] Next, in operation **215**, risk assessment platform **115** evaluates a strength of a match of each possible instance relative to the target entity. In various embodiments, this entails risk assessment platform **115** applying a set of criteria against the indication of the target entity, applying the set of criteria against an indication of a possible instance of the target entity, and comparing results of the application of the sets of criteria of the respective indications. The set of criteria includes sets of weights applicable to the records identified in the indications, such as personal information, geographical information, and occupational information. More specifically, a first set of weights corresponds to a commonness of a surname, and a second set of weights corresponds to a popularity of a domicile. Risk assessment platform **115** may assign a higher value for weights corresponding to popular surnames relative to unpopular surnames. Similarly, risk assessment platform **115** may assign a higher value for weights corresponding to popular, or more populated, countries relative to less populated countries. Risk assessment platform **115** accesses a database to obtain data corresponding to popular surnames and popular countries, then assigns weights to subsets of surnames and countries based on their commonness and popularity, respectively.

[0056] In some embodiments, the set of criteria also includes a third set of weights applicable to personal information missing from an indication of an entity. For example, risk assessment platform **115** assigns a value to a weight corresponding to a missing date of birth and a value to a weight corresponding to a missing gender. In this way, if an indication of an entity is missing either or both of these records, risk assessment platform **115** may take that into account. Other missing information may also be contemplated.

[0057] For the request target entity and each of the possibly related instances, risk assessment platform **115** identifies a surname, domicile, and other identifying information from respective indications. Risk assessment platform **115** applies the first set of weights to the surnames identified and the second set of weights to the domiciles identified to generate results. For any missing pieces of information, such as date of birth and gender, risk assessment platform **115** applies the third set of weights. Then, for each possible instance of the target entity, risk assessment platform **115** compares the results of the application of weights to records associated with the possible instance to the results of the application of weights to records included in the input query for the target entity. Risk assessment platform **115** generates a strength of a match to the target entity for each possible instance based on the comparison results.

[0058] In operation **230**, risk assessment platform **115** displays a visual representation of the possible instances and corresponding strengths of matches on a user interface. In various embodiments, the visual representation of the possible instances illustrates various records associated with the possible instances, such as names, locations, and other identifying information, and the visual representation of the strengths of matches includes a graphical indicator (e.g., a bar, a colored marker) that illustrates a value or range of similarity between a given possible instance and the target entity. In various embodiments, risk assessment platform **115** includes a visualization module configured to obtain the indications of the possible instances and indications of the corresponding strengths of matches and convert such indications to a format readable and executable by a rendering engine of the user device displaying the user interface.

[0059] Optionally, in operation **235**, risk assessment platform **115** identifies whether a strength of a match exceeds or falls below a threshold strength to automate a dismissal of a false positive match, and in turn, reduce the number of false positive instances a user must review advantageously eliminating manual effort and saving the user time. The threshold strength may include a value separating possible instances that have a high probability of being a false positive match from possible instances that have a lower probability of being a false positive match. For a strength of a match exceeding the threshold strength, in operation **240**, risk assessment platform **115** enqueues a corresponding instance for further risk assessment screening. Contrarily, for a strength of a match below the threshold strength, in operation **245**, risk assessment platform **115** dismisses a corresponding instance as being a false positive relative to the target entity.

[0060] In some embodiments, risk assessment platform **115** may include one or more machine learning models or Artificial Intelligence modules trained using training weights and sample enqueuing and dismissal patterns to automate and facilitate false positive detection processes, such as those of method **201**. In such embodiments, risk assessment platform **115** can be iteratively trained with ground truth false positive data to improve detection of false positives based on similarities between records associated with possible instances of a target entity and the target entity. In some such embodiments, a user may initiate re-training or operation of the machine learning models via inputs to the user interface of the user device in communication with risk assessment platform **115** or running an instance of risk assessment platform **115**.

[0061] Referring next to method **202** of FIG. **2**B, in operation **250**, risk assessment platform **115** receives an input (e.g., request **121-1**) from a user. The input includes an indication of a target entity subject to a risk assessment. The indication may include a name of the target entity, an identifier associated with the target entity, a location associated with the target entity, a date of birth of the target entity, and other personal, occupational, and geographic information associated with the target entity. The user may provide the input to risk assessment platform **115** via a user interface of a user device (e.g., user device **120**) in communication with risk assessment platform **115** over a communication network.

[0062] In operation **255**, risk assessment platform **115** identifies possible instances of the target entity specified in the input. To do so, risk assessment platform **115** may first access one or more databases (e.g., database **105**, database **106**, database **116**) and query the database(s) for records matching the indication of the target entity. For example, risk assessment platform **115** performs a query for the target entity using the name, identifier, and location of the target entity. The query results may include multiple entities that share the same or similar names, locations, and other identifying information. Some of these entities may be false positives, while others may correspond to the target entity.

[0063] Next, in operation **260**, risk assessment platform **115** displays a visual representation of the possible instances, corresponding strengths of matches relative to the target entity, and corresponding risk profiles on a user interface. The evaluation processes for determining the strengths of matches between instances and the target entity is described above with respect to method **201** and is excluded here simply for the sake of brevity.

[0064] In various embodiments, the visual representation of the possible instances illustrates various records associated with the possible instances, such as names, locations, and other identifying information, and the visual representation of the strengths of matches includes a graphical indicator (e.g., a bar, a colored marker) that illustrates a value or range of similarity between a given possible instance and the target entity. The visual representation of the risk profiles also illustrates various records associated with the possible instances and additionally illustrates statuses associated with the possible instances, such as an open for review status, a closed review status, a dismissed status relative to a false positive, and more, and chat history associated with the possible instances. In various embodiments, risk assessment platform **115** includes a visualization module configured to obtain the indications of the possible instances, indications of the corresponding strengths of matches, and indications of the risk profiles and convert such indications to a format readable and executable by a rendering engine of the user device displaying the user interface.

[0065] In operation **265**, for a risk profile of a possible instance selected by a user via an input to the user interface, risk assessment platform **115** displays a visual representation of a risk assessment generated for the risk profile including various graphical indicators, textual and numerical elements, and other information. For example, the visual representation of the risk assessment includes an indication of a threat confidence level associated with the selected possible instance. In various embodiments, the threat confidence level corresponds to a level of confidence of corruption with respect to performing financial transactions with the possible instance. In other words, this indicator may correlate to an entity's ability or propensity to engage in unlawful or suspicious activity with financial and non-financial institutions, businesses, or other individuals based on their geographic location, their occupation, their role, their connections to other individuals and entities, and the like.

[0066] To perform the risk assessment and determine the threat confidence level, risk assessment platform **115** accesses one or more databases (e.g., database **105**, database **106**, database **116**) and obtains various records from the databases corresponding to the target entity and the possibly-related instances of the target entity. Risk assessment platform **115** identifies a subset of the obtained records that have values associated therewith from the source databases and converts the values from a first format to a second format to normalize the values to a single format. Risk assessment platform **115** generates a set of weights for the records, including a first set of weights that includes the converted values, and a second set of weights for other records without pre-existing values associated therewith. Then, risk assessment platform **115** applies the sets of weights to respective records to generate the threat confidence level.

[0067] In various embodiments, risk assessment platform **115** performs a comparison between the threat confidence level and a threshold threat level to determine whether the threat confidence level exceeds or falls below the threshold threat level. The threshold threat level may include a value separating low-risk profiles from higher risk profiles, which may require further risk assessment and analysis, such as enhanced due diligence. For a threat confidence level exceeding the threshold threat level, risk assessment platform **115** enqueues a corresponding risk profile for further risk assessment screening. Additionally, risk assessment platform **115** may provide an alert or notification to a user indicative of a high-risk profile. For a threat confidence level below the threshold threat level, risk assessment platform **115** updates the risk profile, or metadata thereof, to indicate that the risk profile is one of a lower risk than one requiring additional screening. Advantageously, automation of high-risk profile flagging and queueing can increase efficiency of initial screening reviews, conserve user time and manual effort, and reduce the probability of missing high-risk entities due to user error, for example. Additionally, the visual indications of high threat confidence levels, as well as notifications thereof, presented on a user interface to a user can improve the user experience and improve efficiency of the screening process.

[0068] In some embodiments, risk assessment platform **115** may include one or more machine learning models or Artificial Intelligence modules trained using training weights and sample threat confidence levels to automate and facilitate risk assessment and analysis processes, such as those of method **202**. In such embodiments, risk assessment platform **115** can be iteratively trained with ground truth confidence threat level data to improve detection of high-risk profiles, as well as low and medium risk profiles for categorization purposes, based on similarities between records having higher risk scores than other records. In some such embodiments, a user may initiate re-training or operation of the machine learning models via inputs to the user interface of the user device in communication with risk assessment platform **115** or running an instance of risk assessment platform **115**.

[0069] FIG. **3** illustrates an example sequence diagram demonstrating access and data flow between elements of a system in accordance with some embodiments of the present technology. FIG. **3** shows sequence **300**, which references elements of operating environment **100** of FIG. **1**. The series of operations shown in sequence **300** may be performed by risk assessment platform **115** of FIG. **1**, and as such, may be implemented in hardware, software, and/or firmware, as well as combinations and variations thereof.

[0070] To begin sequence **300**, risk assessment platform **115** obtains various data sets from multiple databases, such as databases **105**, **106**, and **116**. By way of example, risk assessment platform **115** obtains entity data and common surname data from database **106** and corruption perception index (CPI) data from database **105**. Risk assessment platform **115** additionally, or instead, obtains entity profile data (including entity data, previously generated threat confidence levels, etc.), weights, and other information from database **116**.

[0071] In various embodiments, the CPI data may include a list, table, or other data structure including country names, CPI scores associated with the countries, and ranks of the countries relative to one another. The CPI score may correspond to risk level. The entity data may include a list, a table, or other data structure including a number of individuals or entities, a unique identifier associated with each individual or entity, and a number of records associated with each individual or entity. Examples of the records include a date of birth (applicable for individuals), a gender (applicable for individuals), locations (e.g., place of birth, location of employment, location of residence, location of domicile), countries corresponding to nationality and/or citizenship, a classification or sub-classification, alert types, and the like. More specifically, the classifications may include various categories, or abbreviations thereof, such as non-governmental, national, subnational government, local, international organization, regional organization, state-owned organization, state-invested organization, and instrumentality of state. Some of the classifications may further include sub-classifications, such as close associate entity, close associate, and immediate relative. The alert types may include a number of categories, such as a law enforcement alert, a regulatory enforcement alert, a sanctions alert, a high-risk country alert, and other alerts. Each of the alert types may further include a sub-type, such as a linked entity alert. Using the obtained data, risk assessment platform **115** creates an entity profile for each individual and/or entity in the entity data if one does not already exist in database **116**.

[0072] Risk assessment platform **115** also determines sets of weights and conditions against which to apply the records of a risk profile. For example, risk assessment platform **115** determines a first set of weights applicable to various records of the risk profile to generate a threat confidence level, and a second set of weights applicable to various records of the risk profile to generate a strength of match between a target entity profile and other similar entity records (also referred to as a false positive probability score).

[0073] In various embodiments, each weight of the first set of weights includes a value corresponding to a maximum score achievable if a record matches a condition. The set of conditions may include different threshold levels or direct matches. For example, risk assessment platform **115** may determine a weight for country records having a value of 40. If a risk profile includes the highest-risk country (e.g., based on the CPI data), risk assessment platform **115** may determine that the country record matches a high-risk country condition and award the maximum score for that weight (i.e., 40). If a risk profile includes a country that is not the highest-risk country, risk assessment platform **115** may determine that the country does not match the high-risk country condition but instead matches a different risk condition. As such, risk assessment platform **115** may award a lesser score for that weight. If the condition is a direct match condition, risk assessment platform **115** may award 100% of the weight if the record includes a direct match or 0% of the weight if the record does not include a direct match.

[0074] In various embodiments, each weight of the second set of weights includes values for a direct match of a common last name on one of the common last name lists of the common name data. For example, a weight of the second set of weights may include a value of 40. For a risk profile including a name that matches a name on one of the lists of the common name data, a maximum score of 40 may be determined based on the direct match for the name record.

[0075] In some embodiments, risk assessment platform **115** determines the weights based on a level of corruption or risk associated with the record. For example, risk assessment platform **115** assigns a weight with a high value for a sanctions alert, and a weight with lower value for a law enforcement alert relative to the value of the sanctions alert weight, as sanctions are considered to be more high-risk with respect to financial and commercial dealings and associated potential corruption and unlawfulness. In some such embodiments, risk assessment platform **115** obtains the weights from database **116**. In some such embodiments, risk assessment platform **115** determines the weights based on a request from a user. For example, a user of user device **120** may input custom weight values as part of the request, and risk assessment platform **115** may use the custom weight values for scoring operations and store the custom weights in database **116**.

[0076] In addition to custom weights, a request from user device **120** may identify one or more names and/or unique IDs associated with risk profiles. In response to receiving the request, risk assessment platform **115** identifies risk profile(s) associated with the names or identifiers, and applies the weights and conditions to the records of the risk profiles to generate a threat confidence level and a strength of match of the risk profiles relative to the queried name/identifier.

[0077] By way of example, for a risk profile identified from a search query, risk assessment platform **115** applies weights against the records found in the risk profile and generates the threat confidence level and strength of match for the risk profile. Risk assessment platform **115** may also compare the scores to threshold scores and provide indications of the scores to user device **120** based on either or both score exceeding respective threshold scores. The indications may include a status indicator, a pop-up alert, a message alert, or the like that can be displayed on a user interface of the user device. User device **120** may display the indications as well as an indication of the identified risk profile on a display of user device **120**. A user of user device **120** may take actions based on the threat confidence level, such as reviewing the risk profile, flagging the risk profile for further action, resolving an action associated with the risk profile, and more, based on providing inputs to the user interface. Additionally, or alternatively, risk assessment platform **115** may perform some such actions automatically based on determining a threat confidence level above a respective threshold or a strength of match below a respective threshold.

[0078] FIG. **4** illustrates an example block diagram of a risk assessment platform in accordance with some embodiments of the present technology. FIG. **4** includes operating environment **400**, which includes risk assessment platform **115** and software modules thereof capable of performing risk assessment and analysis, threat detection, and false probability detection with respect to financial dealings with various entities. Risk assessment platform **115** and the software modules thereof may be configured to perform methods thereof, such as methods **200**, **201**, and **202** of FIGS. **2**A, **2**B, and **2**C, respectively. In some embodiments, the modules of risk assessment platform **115** may additionally, or instead, be implemented in hardware and/or firmware, as well as combinations and variations of hardware, firmware, and software.

[0079] In operation, risk assessment platform **115** is configured to perform risk assessment and analysis operations as described above to determine information about various entities, determine a threat confidence level associated with the entities with respect to financial and commercial dealings, and determine how likely prospective entities resulting from an entity search are associated with a target entity that is the query of the search as opposed to a false positive entity. To enable such functionality, risk assessment platform **115** includes several software modules executable by a processing system of risk assessment platform **115**, an example of which includes computing system **1601** of FIG. **16** shown and described below. The software modules include—but are not limited to—input interface module **405**, data processing module **410**, weight generation module **415**, false positive analyzer module **420**, threat detection module **425**, and output interface module **430**.

[0080] Input interface module **405** is representative of a software module configured to receive request **401** including an indication of a target entity subject to a risk assessment, access databases **105**, **106**, and **116** and obtain data corresponding to the target entity based on request **401**, and provide data to other modules of risk assessment platform **115**. The data obtained by input interface module **405** from the databases may include entity records (e.g., identifying information associated with the target entity), weights, criteria, and/or conditions applicable to entity records by other modules, and other risk assessment information.

[0081] Data processing module **410** is representative of a software module configured to identify a subset of the obtained records that have values associated therewith from the source databases and convert the values from a first format to a second format to normalize the values to a single format. Data processing module **410** provides the converted values and corresponding records to weight generation module **415**.

[0082] Weight generation module **415** is representative of a software module configured to generate a set of weights for the records, including a first set of weights that includes the converted values, and a second set of weights for other records without pre-existing values associated therewith. Weight generation module **415** additionally, or instead, receives previously generated weights from a database via input interface module **405**. Then, weight generation module **415** provides the weights to false positive analyzer module **420** and threat detection module **425**.

[0083] False positive analyzer module **420** is representative of a software module configured to apply a subset of the weights to records of several possible instances of the target entity indicated in request **401**. The query results returned from request **401** may include multiple entities that share the same or similar names, locations, and other identifying information. Some of these entities may be false positives, while others may correspond to the target entity. False positive analyzer module **420** evaluates a strength of a match of each possible instance relative to the target entity. In various embodiments, this entails false positive analyzer module **420** applying the subset of weights against the records identified in request **401** for the target entity, applying the subset of weights against records of a possible instance of the target entity, and performing a comparison of the results. A first subset of the weights corresponds to a commonness of a surname, and a second subset of the weights corresponds to a popularity of a domicile. Weight generation module **415** may assign a higher value for weights corresponding to popular surnames relative to unpopular surnames.

[0084] Similarly, weight generation module **415** may assign a higher value for weights corresponding to popular, or more populated, countries relative to less populated countries.

[0085] In some embodiments, the subset of weights also includes a third subset of weights applicable to personal information missing from an indication of an entity. For example, weight generation module **415** assigns a value to a weight corresponding to a missing date of birth and a value to a weight corresponding to a missing gender. In this way, if an indication of an entity is missing either or both of these records, false positive analyzer module **420** may take that into account. Other missing information may also be contemplated.

[0086] For the request target entity and each of the possibly related instances, false positive analyzer module **420** identifies a surname, domicile, and other identifying information from respective indications. False positive analyzer module **420** applies the first subset of weights to the surnames identified and the second subset of weights to the domiciles identified to generate results. For any missing pieces of information, such as date of birth and gender, false positive analyzer module **420** applies the third subset of weights. Then, for each possible instance of the target entity, false positive analyzer module **420** compares the results of the application of weights to records associated with the possible instance to the results of the application of weights to records included in the input query for the target entity. False positive analyzer module **420** generates a strength of a match to the target entity for each possible instance based on the comparison results, and provides the strengths of matches to output interface module **430**.

[0087] Threat detection module **425** is representative of a software module configured to apply a subset of the weights from weight generation module **415** to records associated with the target entity (also collectively referred to as a risk profile of the entity) to generate a threat confidence level for the target entity's risk profile indicative of a likelihood that the entity is or will engage in unlawful financial dealings, such as bribery, money laundering, and the like.

[0088] In various embodiments, to generate the threat confidence level, To perform the risk assessment and determine the threat confidence level, threat detection module **425** applies a subset of weights to the records of the target entity. For example, threat detection module **425** may apply a first weight to a record indicative of the entity's domicile, a second weight to a record indicative of the entity's occupation, a third weight to a record indicative of the entity's political status, a fourth weight to a record indicative of the entity's political, regulatory, law enforcement, and occupational connections, a fifth weight to a record indicative of a presence of sanctions associated with the entity, and the like, as well as combinations and variations thereof based on the records present the risk profile for the entity. Threat detection module **425** combines the results of applying each weight to respective records to determine the threat confidence level and compares the threat confidence level to threshold threat levels to determine a threat classification (e.g., low risk, medium risk, high risk). Threat detection module **425** provides the threat confidence level and the threat classification to output interface module **430**.

[0089] Output interface module **430** is representative of a software module configured to provide indications of the strengths of matches, threat confidence level, the risk profile and records thereof, the possible instances associated with the target entity and records thereof, and the like as output **431** to a user device for instantiation thereof on a user interface. In some embodiments, some such indications may also include alerts indicative of a high threat confidence level to notify a user of the user device as to a high risk entity.

[0090] In some embodiments, one or more modules of risk assessment platform **115**, such as weight generation module **415**, false positive analyzer module **420**, and/or threat detection module **425** may include one or more machine learning models and/or Artificial Intelligence models capable of performing respective functions. More specifically, weight generation module **415** may include a machine learning model trained to generate new weights and corresponding values upon receiving a new record as input. The machine learning model may be trained on training data including sample weights and ground truth values for numerous entries to automate weight generation processes. By way of another example, false positive analyzer module **420** may include a machine learning model trained to identify and dismiss false positive entities to improve efficiency of the screening process and reduce manual effort required of a user. Such a machine learning model or AI model may be trained on training data including sample records and ground truth actions (e.g., confirm, dismiss). By way of yet another example, threat detection module **425** may include a machine learning model trained to identify, classify, and enqueue high-risk entities to improve efficiency of the screening process and reduce manual effort required of a user. Such a machine learning model or AI model may be trained on training data including sample records and ground truth actions (e.g., enqueue, dismiss).

[0091] It may be appreciated that risk assessment platform **115** may include fewer, additional, or different modules capable of performing similar risk assessment and analysis processes. Some capabilities of the modules of risk assessment platform **115** may be performed by different modules.

[0092] FIGS. **5**A, **5**B, and **5**C illustrate aspects of an example user interface associated with a risk assessment platform in accordance with some embodiments of the present technology. More particularly, FIG. **5**A includes aspect **500** of user interface **510**, FIG. **5**B includes aspect **501** of user interface **510**, and FIG. **5**C includes aspect **502** of user interface **510**.

[0093] In various embodiments, user interface **510** is representative of a graphical user interface of a user device running or in communication with a risk assessment platform, such as risk assessment platform **115**. User interface **510** may include various dashboards, toolbars, search bars, text boxes, icons, drop-downs, indications, and the like, with which a user of the user device may view, click, edit, or otherwise interact with the risk assessment platform. For example, user interface **510** includes account toolbar **511**, functional toolbar **512**, and a dashboard component capable of displaying different content based on an option selected via one of the elements of account toolbar **511** and functional toolbar **512**. Account toolbar **511** may include a set of icons related to settings, help, account information, notifications, or other account-related functions. Functional toolbar **512** may include a set of icons related to different dashboards navigable on an application.

[0094] An example dashboard viewable via an element of functional toolbar **512** includes name screening dashboard **513** as shown in aspect **500** of FIG. **5**A. Name screening dashboard **513** includes visual and textual indications corresponding to entity search **515** whereby a user can enter information about a target entity and search for the target entity according to the entered records via search button **516**. Examples of the records enterable by a user via entity search **515** instantiated on name screening dashboard **513** include a name, a user identifier (UID), a date of birth, a gender, known geographical locations, and other information (e.g., known aliases, known connections, occupation). Additional, fewer, or different fields may be contemplated.

[0095] Upon submission of one or more of the records in entity search **515** via search button **516**, the risk assessment platform accesses one or more databases to identify possible instances associated with the records and the target entity. The results obtained by the risk assessment platform in response to the entity search **515** may include multiple entities that share the same or similar names, locations, and other identifying information. Some of these entities may be false positives, while others may correspond to the target entity.

[0096] The risk assessment platform evaluates a strength of a match of each possible instance relative to the target entity. This may entail applying a set of criteria against the indication of the target entity, applying the set of criteria against an indication of a possible instance of the target entity, and comparing results of the application of the sets of criteria of the respective indications. For each of the possibly related instances, the risk assessment platform generates a strength of a match to the target entity and displays indications of the strengths of matches on name screening dashboard **513**, as shown in aspect **501** of FIG. **5**B.

[0097] Referring more specifically to aspect **501** of FIG. **5**B, in this aspect, user interface **510** includes entity status window **520**, risk assessment window **525**, false positive identification window **530**, and history window **535**.

[0098] Entity status window **520** includes various indications related to status of a risk profile for a target entity queried via entity search **515**, status of a review queue including possible instances related to the risk profile, and other status indicators. Entity status window **521** also includes a re-screen element **521** by which a user can refresh the information presented in entity status window **520** and risk assessment window **525**, among other elements of name screening dashboard **513**. Upon submission of a refresh by a user, the risk assessment platform may perform a new search for the target entity by accessing various databases and identifying associated records, and any updates thereto, for presentation on name screening dashboard **513**.

[0099] Risk assessment window **525** includes various tabs that include indications of information associated with the risk profile of the target entity. As shown in aspect **501**, user interface **510** shows profile view **526** on risk assessment window **525**. Other tabs or views may be navigable by a user triggering the rendering and displaying of other information about the target entity.

[0100] Profile view **526** includes personal and geographical information about the risk profile of the target entity, such as a list or queue among which the risk profile belongs, a unique identifier of the risk profile, a name associated with the risk profile, a date on which the risk profile was created, a date on which the risk profile was last updated, a category to which the entity belongs, and country locations, among other information.

[0101] False positive identification window **530** may include indications of one or more possible instances of risk profiles related to the target entity, text descriptions about the instances, selection elements to view or interact with one or more of the instances profiles, and visualizations of strength of match to the target entity, such as strength of match indicator **532**. In the example illustrated in aspect **501**, false positive identification window **530** shows three risk profiles associated with the name "John Doe". The first instance in the list may include strength of match indicator **532** indicative of an exact match to parameters requested by a user via entity search **515**, which may be determined based on a false positive probability analysis performed by the risk assessment platform (e.g., method **201** of FIG. **2**A). The second instance in the list may include a strength of match indicative of a strong match to parameters requested by the user. The third instance in the list may include a strength of match indicative of a medium match to parameters requested by the user.

[0102] The values of the strengths of matches, and corresponding indicators, may be determined by the risk assessment platform in response to the user requesting a search for the target entity. For example, upon inputting a search via search button **516**, the risk assessment platform can identify the possible instances by comparing the records entered by the user to records obtained by the risk assessment platform from various databases. The risk assessment platform evaluates the possible instances against the target entity to determine a strength of match for each possible instance relative to the target entity. For a possible instance including many similar records as the ones entered by the user, the risk assessment platform may visualize the strength of match as an exact match as shown by strength of match indicator **532**. For a possible instance including some similar records as the ones entered by the user, the risk assessment platform may visualize the strength of match as a strong match. For a possible instance including fewer records than a strong match, the risk assessment platform may visualize the strength of match as a medium match, and so on.

[0103] During a screening process, a user can view details about the possible instances via a selection element of false positive identification window **530**. For any selected instances, a user can confirm the instance(s) as a match using confirm button **531** or dismiss the instance(s) as a false positive using dismiss button **532**. Upon confirmation of an instance, the risk assessment platform may merge records of the instance with the target entity's risk profile. Upon dismissal of an instance, the risk assessment platform may update metadata associated with the instance to indicate that the instance is a false positive with respect to the target entity. Additionally, or alternatively, the risk assessment platform may remove the instance from a queue including records for screening.

[0104] History window **535** includes indications of previous activity related to the target entity, including status updates, textual updates entered or received by the user, and the like.

[0105] Moving next to aspect **502** of FIG. **5**C, aspect **502** shows report view **527** within risk assessment window **525** of name screening dashboard **513**. To navigate to report view **527** from another view of risk assessment window **525**, a user may click, tap, or otherwise interact with the indication labeled "report" on user interface **510**. In response to navigating to report view **527**, the risk assessment platform performs a risk assessment on the target entity to populate report view **527** with information about the target entity including threat confidence level **528**. Information displayed in report view **527** may include a name, affiliated countries, a classification, known alerts, sanctions, and connections associated with the target entity. Threat confidence level **528** includes a visual indication, such as a meter with a sliding scale indicator **529** as shown in aspect **502**, indicative of a likelihood the target entity is or will engage in unlawful activity with respect to financial and commercial dealings.

[0106] In various embodiments, to perform the risk assessment and calculate threat confidence level **528**, the risk assessment platform accesses the various databases and obtains records associated with the target entity, then applies weights to the records to generate threat confidence level **528**. Details of such operations are discussed with respect to method **202** of FIG. **2**B, and thus, are excluded here simply for the sake of brevity.

[0107] As shown in aspect **502**, for the target entity, John Doe, the risk assessment platform determines a threat confidence level **528** having a high score as shown by sliding scale indicator **529**. In various embodiments, for a high-risk entity, the risk assessment platform may enqueue the entity for further risk assessment and analysis. For lower-risk entities (e.g., entities having a threat confidence level **528** below a threshold threat level), the risk assessment platform might not add the entities to such a queue.

[0108] FIG. **6** illustrates another aspect of user interface **510** in accordance with some embodiments of the present technology. FIG. **6** shows aspect **600**, which shows false positive analysis dashboard **610** of user interface **510**.

[0109] False positive analysis dashboard **610** is representative of a view instantiated on user interface **510** by the risk assessment platform upon an input to functional toolbar **512**. Upon an input to navigate to false positive analysis dashboard **610**, the risk assessment platform generates various metrics related to false positives present in a user's data and displays graphical indications visualizing the corresponding metrics, such as visualizations **611**, **612**, **613**, **614**, **615**, and **616**.

[0110] For example, the risk assessment platform can determine a first set of metrics indicative of an entity type deriving the most false positive instances in the user's data. The risk assessment platform may determine the first set of metrics by applying false positive detection processes (e.g., method **201** of FIG. **2**A) for numerous combinations and variations of target entities and determining a number of possible instances having strengths of matches below a threshold level (i.e., false positive instances). Then, the risk assessment platform may compute which searched entity type returns the most false positive instances. Based on the results, the risk assessment platform displays visualization **611** to convey the results in a graphical manner to the user on user interface **510**.

[0111] The risk assessment platform may perform additional false positive analysis on a user's data to determine other sets of metrics and generate visualizations for the other sets of metrics. By way of another example, the risk assessment platform can determine a second set of metrics indicative a data set within a user's data (e.g., PEP classification data sets separated out by different PEP classification) that generates the most false positive instances. The risk assessment platform displays visualization **612** corresponding to results of a false positive detection process with respect to the second set of metrics. By way of another example, the risk assessment platform can determine a third set of metrics indicative a country within a user's data that generates the most false positive instances. The risk assessment platform displays visualization **613** corresponding to results of a false positive detection process with respect to the third set of metrics. The risk assessment platform may perform such analysis for a number of different metrics based on inputs to the risk assessment platform.

[0112] Advantageously, the risk assessment platform can detect problematic records in a user's data set yielding high numbers of false positives. False positives can increase manual effort as more records must be screened by a user, which reduces efficiency of the screening process, reduces storage capacity within databases, and reduces processing capacity as additional, unnecessary records may be returned during queries.

[0113] Additionally, user interface **510** enhances the user experience and screening process as the visualizations of information on user interface **510** can promote efficiency, ease of use, and convenience for users who may be required to screen hundreds or thousands of entities a day.

[0114] The following figures, FIGS. **7**, **8**, **9**, **10**A, **10**B, **11**A, **11**B, **12**A, **12**B, **13**, **14**A, **14**B, **14**C, **14**D, and **15** illustrate example aspects related to risk assessments and analysis of high-risk individuals and entities, such as politically-exposed persons (PEPs). More particularly, the following figures and related discussion correspond to specific embodiments involving risk and governance data, corruption perception data, linked entity data, and the like, as well as configurable sets of weights and values associated therewith, applicable to performing risk assessments and analyses on politically-exposed persons (PEPs). Other data, records, values, and weights may be contemplated in this context, as well as other contexts. Specifically, FIG. **7** relates to categories/classifications and associated weights configurable in a user interface (e.g., UI **510**) for generation and instantiation of a risk assessment on the user interface as shown and described above, while FIGS. **8-15** includes tables with configurable information used in such risk assessments and analyses.

[0115] Generally, a PEP refers to an individual or an entity capable of being influenced financially based on a political status or political connection. Various institutions and companies track PEPs and compile data associated with PEPs, such as name, citizenship, domicile, residence, national origin, ethnicity, race, occupation, associations or connections, and the like as part of enhanced due diligence requirements. PEPs are screened by financial institutions and non-financial institutions and companies to determine levels of risk associated with the PEPs with respect to financially transacting with the PEPs.

[0116] Problematically, PEP data is often duplicative, overlapping, and unfilterable, which can lead to high amounts of false positives and significant manual effort on behalf of users (e.g., financial institution employees, cybersecurity and risk experts) to review. As disclosed herein, risk assessment techniques may be employed to generate risk assessments and analyses, including various confidence level scores, by applying configurable and dynamic weights against different records collected about entities, such as PEPs. Results of the assessments may be filtered and compared against threshold levels to provide condensed queues of risk profiles for review by a user. Advantageously, such techniques may not only reduce the number of PEPs in a review queue but also increase efficiency and accuracy when reviewing PEPs, as the number of false positives may be reduced (or at least visually indicated in a user interface, for example). Additionally, visual indicators, notifications, and alerts corresponding to high-risk PEPs may be presented in a user interface to facilitate review of PEPs and organize information typically spread out across several databases into an integrated dashboard view.

[0117] Referring now to FIG. **7**, FIG. **7** illustrates an aspect of an example user interface associated with a risk assessment platform in accordance with some embodiments of the present technology. FIG. **7** shows user interface **510**, which displays various elements and indications related to risk assessments.

[0118] In various embodiments, user interface **510** includes various dashboards, toolbars, search bars, text boxes, icons, drop-downs, indications, and the like, with which a user may view, click, edit, or otherwise interact. For example, user interface **510** includes account toolbar **511** and functional toolbar **512**. User interface **510** also includes a dashboard component on which different content, based on an option selected via one of the elements of account toolbar **511** or functional toolbar **512**, is rendered and displayed to the user. Account toolbar **511** may include a set of icons related to settings, help, account information, notifications, or other account-related functions. Functional toolbar **512** may include a set of icons related to different dashboards navigable on an application, such as a screening dashboard, risk assessment and analysis dashboard, and the like.

[0119] An example dashboard view selectable via functional toolbar **512** includes PEP risk scoring configuration **710** as shown in aspect **700** of FIG. **7**. PEP risk scoring configuration **710** includes PEP classification list **711**, weight configuration list **713**, and a save component **715**. PEP classification list **711** includes classifications **712** corresponding to a list of classifications defined by one or more organizations (e.g., World Check, Dow-Jones, Rzolut). In aspect **700**, classifications **1114** includes the following classifications: PEP non-governmental, PEP non-governmental—close associate entity, PEP non-governmental—close associate, PEP non-governmental—immediate relative, PEP national, PEP national—close associate entity, PEP national—close associate, PEP national—immediate relative, PEP local, PEP local—close associate entity, PEP local—close associate, PEP local—immediate relative, PEP international organization, PEP international organization—close associate entity, PEP international organization—close associate, and PEP international organization—immediate relative. In some embodiments, additional, fewer, or different classifications, as well as combinations and variations thereof, may be contemplated.

[0120] Each classification may have a corresponding menu, one of configuration menus **714**, by which a user can configure a weight assigned to a classification for risk assessment and analysis purposes. Examples of the configuration menus **714** may include drop-down menus, radio buttons, text boxes, and the like. In the example shown in aspect **700**, configuration menus **714** include drop-down menus that each include options like high, medium, and low, which correspond to weight ranges or thresholds. By way of example, a high option selected for a classification may correspond to a higher, or more valuable, weight when applying the weights to records found for an entity. In some embodiments, a user may be able to enter in a custom value via the dashboard component of user interface **510** instead of, or in addition to, a selection via configuration menus **714**.

[0121] In some embodiments, a user may select from default settings or may input weight ranges via configuration menus **714** for use by a risk assessment platform (e.g., risk assessment platform **115** of FIG. **1**) in risk-based level computations. After selecting the weight ranges for each classification of PEP classification list **711**, a user may save the selections via save component **715** of user interface **510**. The selections may be saved to a database or memory location accessible by the risk assessment platform (e.g., database **116**) via an interface coupling the risk assessment platform, the user device instantiating user interface **510**, and the destination storage location.

[0122] Referring next to the tables illustrating example weight values and corresponding record classifications/categories, FIG. **8** illustrates example scoring tables and factors used in an implementation. FIG. **8** shows tables **801**, **802**, and **803**, which may be used by a risk assessment, threat computation and false positive analysis system (e.g., risk assessment platform **115** of FIG. **1**) to perform risk assessment and analysis techniques (e.g., methods **201** and **202** of FIGS. **2**A and **2**B, respectively).

[0123] Table **801** includes threshold ranges that may be used to categorize risk levels for a risk assessment in an embodiment. Table **801** includes range **810** and score **811**. Range **810** includes a low range, a medium range, and a high range. The low range may include a score **811** between 0 and 54, the medium range may include a score **811** between 55 and 75, and the high range may include a score **811** of 76 and higher. Other ranges and scores may also be contemplated and used in risk assessment operations described herein.

[0124] Table **802** includes factors that may be considered when scoring individual or entity profiles (e.g., PEP profiles) and records thereof. Table **802** includes number **815** and factor **816**. Number 1 corresponds to a first factor **816** titled Customer & Entity, and number 2 corresponds to a second factor **816** entitled "Geographic Location." Other factors may also be contemplated and used in scoring operations described herein.

[0125] Table **803** shows a list of record categories **821** with associated IDs **820** and weights **822**. The list may also include an indication of an override factor **823** that may be used to influence a risk level during a risk assessment operation. Number 1 under ID **820** may correspond to a record category titled Country CPI (where CPI stands for corruption perception index) that has a weight with a maximum value of 40. A record of a PEP profile that indicates one or more countries may be given a value between 0 and 40 based on CPI data obtained from a database (e.g., database **105**). Number 2 under ID **820** may correspond to a record category titled PEP Classification that has a weight with a maximum value of 40. A record of a PEP profile that indicates a classification and/or sub-classification of the PEP (e.g., non-governmental, national, local) may be given a value between 0 and 30 based on PEP data obtained from a database (e.g., database **106**). Number 3a under ID **820** may correspond to a record category titled Law Enforcement Alert that has a weight with a maximum value of 30. A record of a PEP profile that indicates a classification and/or sub-classification of the PEP corresponding to law enforcement may be given a value between 0 and 30 based on the obtained PEP data. Number 3b under ID **820** may correspond to a record category titled Regulatory Enforcement Alert that has a weight with a maximum value of 30. A record of a PEP profile that indicates a classification and/or sub-classification of the PEP corresponding to regulatory enforcement may be given a value between 0 and 30 based on the obtained PEP data. Number 3c under ID **820** may correspond to a record category titled Other Bodies Alert that has a weight with a maximum value of 30. A record of a PEP profile that indicates a classification and/or sub-classification of the PEP corresponding to other governmental bodies, agencies, or the like may be given a value between 0 and 30 based on the obtained PEP data. Number 3d under ID **820** may correspond to a record category titled Sanctions Alert that has a weight with a maximum value of 76. A record of a PEP profile that indicates a classification and/or sub-classification of the PEP corresponding to a sanction (e.g., a legal sanction) may be given a value between 0 and 76 based on the obtained PEP data.

[0126] As illustrated in table **803**, override **823** may be indicated as "Yes" in association with ID **3***d* as a PEP having a record that indicates a sanction may be deemed as very high risk relative to other record categories **821**. In some embodiments, the value of weight **822** for a record with override **823** indicated as "Yes" may be automatically set to meet or exceed the high range **810**. Accordingly, the value of weight **822** may equal the minimum score **811** of table **801**, such that when a PEP profile includes an indication of a record flagged with override **823**, a scoring system may indicate the PEP profile as high risk. Number 4a under ID **820** may correspond to a record category titled Law Enforcement Alert (Linked Entity) that has a weight with a maximum value of 20. A linked entity may refer to a connection, either an individual or entity, of the person or entity identified in the PEP profile, such as a business associate or partner, a family relative, or the like. Thus, a record of a PEP profile that indicates a linked entity involved in a law enforcement agency may be given a value between 0 and 20 based on the obtained PEP data. Number 4b under ID **820** may correspond to a record category titled Regulatory Enforcement Alert (Linked Entity) that has a weight with a maximum value of 20. A record of a PEP profile that indicates a linked entity involved in a regulatory enforcement agency may be given a value between 0 and 20 based on the obtained PEP data. Number 4c under ID **820** may correspond to a record category titled Sanctions Alert (Linked Entity) that has a weight with a maximum value of 20. A record of a PEP profile that indicates a linked entity that has been sanctioned may be given a value between 0 and 46 based on the obtained PEP data. Number 4d under ID **820** may correspond to a record category titled PEP Classification (Linked Entity) that has a weight with a maximum value of 20. A record of a PEP profile that indicates a classification or sub-classification of a linked entity may be given a value between 0 and 20 based on the obtained PEP data. Number 5 under ID **820** may correspond to a record category titled High Risk Countries that has a weight with a maximum value of 20. A record of a PEP profile that indicates a high risk country (e.g., Russia, China) may be given a value between 0 and 20 based on the obtained PEP data.

[0127] The following Figures, FIGS. **9**, **10**A-B, **11**A-B, **12**A-B, **13**, and **14**A-D, illustrate and describe further details related to the aforementioned IDs, record categories, and weights as described in FIG. **8**.

[0128] FIG. **9** illustrates example scoring tables and factors used in an implementation.

[0129] FIG. **9** shows tables **901** and **902**, which may be used by a scoring system (e.g., risk assessment platform **115**) to perform risk-based scoring techniques (e.g., method **201**, method **202**).

[0130] Table **901** shows a list of parameters associated with the record category labeled number 1 under ID **820** of table **803** of FIG. **8**. Accordingly, table **901** refers to record category **821** titled Country CPI. Country CPI may include a value of 1 for ID **820**, a value of 2 for factor number **815** (i.e., a geographic location), an indication of no override **823**, a value of 40 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "direct". Module **917** may refer to a condition type. In some embodiments, a direct condition type refers to a direct scoring method, meaning the value of weight **822** may be calculated based on a record found in the PEP profile and the Country CPI rank.

[0131] Table **902** shows a sample list of countries under country name **920**, a respective CPI score **921**, a respective rank **922**, and a respective normalized score **923**. As illustrated, table **902** includes Somalia under country name **920**, which may have the highest rank **922** and normalized score **923**, based on rank **922**, and the lowest CPI score **921**. Based on these numbers, a PEP profile that includes an indication of Somalia (i.e., a direct match) may receive the maximum value of weight **822** (40). Table **902** also includes Denmark under country **920**, which may have the lowest rank **922** and normalized score **923**, based on rank **922**, and the lowest CPI score **921**. Based on these numbers, a PEP profile that includes an indication of Denmark may receive the lowest value of weight **822** (0). Normalized score **923** may include a scaled value corresponding to a percentage of weight **822**. In some embodiments, a scoring system may use normalized score **923** and the value of weight **822** to determine a score for Country CPI. To produce normalized score **923**, a risk assessment platform (e.g., risk assessment platform **115**) may identify values associated with country records from a database (i.e., CPI score **921**) and convert the values from their original format to a different format standardized relative to other records and values obtained by the risk assessment platform. In this way, the risk assessment platform can apply weights to records in a standardized manner to simply risk level computations and false positive probability computations.

[0132] FIGS. **10**A and **10**B illustrate example scoring tables and factors used in an implementation. FIG. **10**A shows table **1001**, and FIG. **10**B shows table **1002**, both of which may be used by a scoring system (e.g., risk assessment platform **115**) to perform risk-based scoring techniques (e.g., method **201**, method **202**).

[0133] Table **1001** of FIG. **10**A shows a list of parameters associated with the record category labeled number 2 under ID **820** of table **803** of FIG. **8**. Accordingly, table **1001** refers to record category **821** titled PEP Classification. PEP Classification may include a value of 2 for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 30 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match". In some embodiments, a match condition type refers to a matching scoring method, meaning the value of weight **822** may be calculated based on a direct match between a record found in the PEP profile and a PEP classification condition table (table **1002**).

[0134] Table **1002** of FIG. **10**B shows a list of PEP classifications and associated scores. Table **1002** includes a first group of PEP classifications in the first row under the first column of table **1002** titled condition threshold **1010**. The PEP classifications included in the first group may include non-governmental PEPs, national PEPs, subnational PEPs, and more. This first group may correspond to score **1011** of 100. In other words, if a PEP profile includes an indication of a PEP classification in the first group, a scoring system may award 100% of the value of weight **822** (i.e., 30) based on module **917**. Table **1002** also includes a second group of PEP classifications in the second row under condition threshold **1010**. The PEP classifications included in the second group may include close associate linked entities, international organizations, regional organizations, and more. The second group may correspond to score **1011** of 67. In other words, if a PEP profile includes an indication of a PEP classification in the second group, a scoring system may award 67% of the value of weight **822** (i.e., 20). Table **1002** also includes a third group of PEP classifications in the third row under condition threshold **1010**. The PEP classifications included in the third group may include state-owned organizations, state-invested organizations, and instrumentalities of a state, and more. The third group may correspond to score **1011** of 33. In other words, if a PEP profile includes an indication of a PEP classification in the third group, a scoring system may award 33% of the value of weight **822** (i.e., 10). Table **1002** further includes a fourth group of PEP classifications in the fourth row under condition threshold **1010**. The PEP classifications included in the fourth group may include immediate relatives of various classifications. The fourth group may correspond to score **1011** of 0. In other words, if a PEP profile includes an indication of a PEP classification in the fourth group, a scoring system might not award any score of weight **822**. Other groupings and score breakdowns may be contemplated.

[0135] FIGS. **11**A and **11**B illustrate example scoring tables and factors used in an implementation. FIG. **11**A shows tables **1101** and **1102**, and FIG. **11**B shows table **1103** and **1104**, each of which may be used by a scoring system (e.g., risk assessment platform **115**) to perform risk-based scoring techniques (e.g., method **201**, method **202**).

[0136] Table **1101** of FIG. **11**A shows a list of parameters associated with the record category labeled number 3a under ID **820** of table **803** of FIG. **8**. Accordingly, table **1101** refers to record category **821** titled Law Enforcement Alert. This category may include a value of 3a for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 20 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0137] Table **1102** of FIG. **11**A shows a list of parameters associated with the record category labeled number 3b under ID **820** of table **803** of FIG. **8**. Accordingly, table **1102** refers to record category **821** titled Regulatory Enforcement Alert. This category may include a value of 3b for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 20 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0138] Table **1103** of FIG. **11**B shows a list of parameters associated with the record category labeled number 3c under ID **820** of table **803** of FIG. **8**. Accordingly, table **1103** refers to record category **821** titled Other Bodies Alert. This category may include a value of 3c for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 20 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0139] Table **1104** of FIG. **11**B shows a list of parameters associated with the record category labeled number 3d under ID **820** of table **803** of FIG. **8**. Accordingly, table **1104** refers to record category **821** titled Sanctions Alert. This category may include a value of 3d for ID **820**, a value of 2 for factor number **815**, an indication of override **823**, a value of 76 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0140] FIGS. **12**A and **12**B illustrate example scoring tables and factors used in an implementation. FIG. **12**A shows tables **1201** and **1202**, and FIG. **12**B shows table **1203** and **1204**, each of which may be used by a scoring system (e.g., risk assessment platform **115**) to perform risk-based scoring techniques (e.g., method **201**, method **202**).

[0141] Table **1201** of FIG. **12**A shows a list of parameters associated with the record category labeled number 4a under ID **820** of table **803** of FIG. **8**. Accordingly, table **1201** refers to record category **821** titled Law Enforcement Linked Entity Alert. This category may include a value of 4a for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 20 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0142] Table **1202** of FIG. **12**A shows a list of parameters associated with the record category labeled number 4b under ID **820** of table **803** of FIG. **8**. Accordingly, table **1202** refers to record category **821** titled Regulatory Enforcement Linked Entity Alert. This category may include a value of 4b for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 20 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0143] Table **1203** of FIG. **12**B shows a list of parameters associated with the record category labeled number 4c under ID **820** of table **803** of FIG. **8**. Accordingly, table **1203** refers to record category **821** titled Sanctions Linked Entity Alert. This category may include a value of 4c for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 46 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0144] Table **1204** of FIG. **12**B shows a list of parameters associated with the record category labeled number 4d under ID **820** of table **803** of FIG. **8**. Accordingly, table **1204** refers to record category **821** titled PEP Classification. This category may include a value of 4d for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 76 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0145] FIG. **13** illustrates example scoring tables and factors used in an implementation. FIG. **13** shows tables **1301** and **1302**, which may be used by a scoring system (e.g., risk assessment platform **115**) to perform risk-based scoring techniques (e.g., method **201**, method **202**).

[0146] Table **1301** of FIG. **13** shows a list of parameters associated with the record category labeled number 5 under ID **820** of table **803** of FIG. **8**. Accordingly, table **1301** refers to record category **821** titled High Risk Countries. This category may include a value of 5 for ID **820**, a value of 2 for factor number **815**, an indication of no override **823**, a value of 20 for weight **822**, a value of 0 for delay duration of risk **915**, a value of 0 for duration of risk **916**, and a module **917** type of "match".

[0147] Table **1302** of FIG. **13** shows a list of high risk countries and associated scores. Table **1302** includes a first group of high risk countries in the first row under the first column of table **1302** titled condition threshold **1310**. The first group may include the Russian Federation, among other countries. This first group may correspond to score **1311** of 100. In other words, if a PEP profile includes an indication of the Russian Federation, a scoring system may award 100% of the value of weight **822** (i.e., 20) based on module **917**. Table **1302** includes a second group of high risk countries in the second row under the first column of table **1302** titled condition threshold **1310**. The second group may include China, among other countries. This second group may correspond to score **1311** of 50. In other words, if a PEP profile includes an indication of China, a scoring system may award 50% of the value of weight **822** (i.e., 10) based on module **917**. Other countries and scores may be contemplated.

[0148] The following tables in FIGS. **14**A, **14**B, **14**C, and **14**D demonstrate examples of identified records in a sample PEP profile and corresponding scores based on weights **822** and modules **917** as applied against the records.

[0149] FIGS. **14**A, **14**B, **14**C, and **14**D illustrate example scoring tables and factors used in an implementation. FIG. **14**A shows table **1401**, FIG. **14**B shows table **1402**, FIG. **14**C shows table **1403**, and FIG. **14**D shows table **1404**, which may include scoring results produced by a scoring system (e.g., risk assessment platform **115**) upon performing risk-based scoring techniques (e.g., method **201**, method **202**).

[0150] Table **1401** of FIG. **14**A shows a first sample risk-based score for a first sample PEP profile. Table **1401** includes various factors **1410**, breakdowns of factor score **1411** for each factor or record, breakdowns of weight score **1412** for each factor or record, and a total score **1418** that includes a combination of each weight score **1412**. Factors **1410** may include countries **1413**, PEP classification **1414**, alerts **1415**, linked entities **1416**, and high-risk countries **1417**.

[0151] In this example, the first sample PEP profile may include Iraq, Jordan, Syria, and United Arab Emirates under countries **1413**. More specifically, Iraq may be identified as a country with a factor score **1411** of 86%, Jordan may be identified as a country with a factor score **1411** of 55%, Syria may be identified as a country with a factor score **1411** of 99%, and UAE may be identified as a country with a factor score **1411** of 29%. Based on these countries **1413**, a scoring system may determine that the countries correspond to a factor score **1411** of 99% (the highest of any identified country). The scoring system may multiply the factor score **1411** with the maximum value of the weight (e.g., 40) to obtain a weight score **1412** of 39.6.

[0152] The PEP profile may include a national classification under PEP classification **1414**. Based on a direct match to a classification in the first group of PEP classifications in table **602** of FIG. **6**B, the scoring system may determine that the classification corresponds to a factor score **1411** of 100%, and thus, the full value of the weight score **1412** may be determined (e.g., 30) for this factor **1410**.

[0153] The PEP profile may also include several indications corresponding to alerts **1415**, such as a sanctions alert, a regulatory enforcement alert, a law enforcement alert, and an other bodies alert. Based on these indications and a match to an alert type, the scoring system may determine that the alerts each correspond to a factor score **1411** of 100%, and thus, the full value of each weight score **1412** may be determined. In this example, the sanctions alert may be flagged as an override factor and include a higher weight score **1412** than other factors.

[0154] The PEP profile may include several indications corresponding to linked entities **1416**, such as a PEP classification linked entity, a sanctioned linked entity, a regulatory enforcement linked entity, and a law enforcement linked entity. Thus, the scoring system may determine that the linked entities correspond to a factor score **1411** of 100%, and the scoring system may award the full value of each weight score **1412** for each of the records under linked entities **1416**.

[0155] Lastly, the PEP profile may include an indication of high-risk countries **1417**, which may include Syria. Syria may be included in a first group of high-risk countries, which may correspond to a factor score **1411** of 100%, and thus, the scoring system may award the full value of weight score **1412** for the indication of high-risk countries **1417**.

[0156] Based on factors **1410** and the weight score **1412** for each factor identified in the first sample PEP profile, the scoring system may calculate a total score **1418** of 285.6. The scoring system may compare total score **1418** to threshold scores, such as scores **411** of table **401** of FIG. **4**, to determine whether the PEP is a high-risk PEP, a medium-risk PEP, or a low-risk PEP. Because total score **1418** exceeds the high range, the PEP of the first sample PEP profile may be identified as high-risk.

[0157] Table **1402** of FIG. **14**B shows a second sample risk-based score for a second sample PEP profile. In this example, the second sample PEP profile may include Indonesia (a 72% factor score **1411**) under countries **1413**. Based on the countries **1413** and the percentage match thereof, a scoring system may determine that the countries correspond to a factor score **1411** of 72%. The scoring system may multiply the factor score **1411** by the maximum value of the weight (e.g., 40) to obtain a weight score **1412** of 28.8.

[0158] The PEP profile may include a local government immediate relative classification (PEP L-R) under PEP classification **1414**. Based on a direct match to a classification in the second group of PEP classifications in table **1002** of FIG. **10**B, the scoring system may determine that the classification corresponds to a factor score **1411** of 67%, and thus, 67% of the weight score **1412** may be determined (e.g., 20) for this factor **1410**.

[0159] The PEP profile might not include any alerts **1415** or high-risk countries **1417**.

[0160] However, the PEP profile may include an indication of a PEP classification under linked entities **1416**, such that the scoring system may award 100% of the weight associated with linked entities **1416** (e.g., 20).

[0161] Based on factors **1410** and the weight score **1412** for each factor identified in the second sample PEP profile, the scoring system may calculate a total score **1418** of 68.8. The scoring system may compare total score **1418** to threshold scores, such as scores **411** of table **401** of FIG. **4**, to determine whether the PEP is a high-risk PEP, a medium-risk PEP, or a low-risk PEP. Because total score **1418** exceeds the minimum score of the medium range but does not exceed the minimum score of the high range, the PEP of the second sample PEP profile may be identified as medium-risk.

[0162] Table **1403** of FIG. **14**C shows a third sample risk-based score for a third sample PEP profile. In this example, the third sample PEP profile may include Denmark under countries **1413**. Based on the countries **1413** and the percentage match thereof, a scoring system may determine that the countries correspond to a factor score **1411** of 0%. The scoring system may multiply the factor score **1411** by the maximum value of the weight (e.g., 40) to obtain a weight score **1412** of 0.

[0163] The PEP profile may include a subnational government immediate relative entity classification (PEP SN-R) under PEP classification **1414**. Based on a direct match to a classification in the fourth group of PEP classifications in table **1002** of FIG. **10**B, the scoring system may determine that the classification corresponds to a factor score **1411** of 0%, and thus, 0% of the weight score **1412** may be determined (e.g., 0) for this factor **1410**.

[0164] The PEP profile might not include any alerts **1415** or high-risk countries **1417**.

[0165] However, the PEP profile may include an indication of a PEP classification under linked entities **1416**, such that the scoring system may award 100% of the weight associated with linked entities **1416** (e.g., 20).

[0166] Based on factors **1410** and the weight score **1412** for each factor identified in the second sample PEP profile, the scoring system may calculate a total score **1418** of 20. The scoring system may compare total score **1418** to threshold scores, such as scores **411** of table **401** of FIG. **4**, to determine whether the PEP is a high-risk PEP, a medium-risk PEP, or a low-risk PEP. Because total score **1418** exceeds the minimum score of the low range but does not exceed the minimum score of the medium range, the PEP of the third sample PEP profile may be identified as low-risk.

[0167] Table **1404** of FIG. **14**D shows a fourth sample risk-based score for a fourth sample PEP profile. In this example, the fourth sample PEP profile may include Ghana under countries **1413**. Based on the countries **1413** and the percentage match thereof, a scoring system may determine that the countries correspond to a factor score **1411** of 60%. The scoring system may multiply the factor score **1411** by the maximum value of the weight (e.g., 40) to obtain a weight score **1412** of 24.

[0168] The PEP profile may include a local government classification (PEP L) under PEP classification **1414**. Based on a direct match to a classification in the second group of PEP classifications in table **602** of FIG. **6**B, the scoring system may determine that the classification corresponds to a factor score **1411** of 67%, and thus, 67% of the weight score **1412** may be determined (e.g., 20) for this factor **1410**. The PEP profile might not include any alerts **1415**, linked entities **1416**, or high-risk countries **1417**.

[0169] Based on factors **1410** and the weight score **1412** for each factor identified in the second sample PEP profile, the scoring system may calculate a total score **1418** of 44. The scoring system may compare total score **1418** to threshold scores, such as scores **411** of table **401** of FIG. **4**, to determine whether the PEP is a high-risk PEP, a medium-risk PEP, or a low-risk PEP. Because total score **1418** exceeds the minimum score of the medium range but does not exceed the minimum score of the high range, the PEP of the fourth sample PEP profile may be identified as medium-risk. However, the PEP profile may further include a factor **1410** indicative that the PEP is deceased, which may be referred to as deceased override **1419**. Because the PEP is deceased, a weight score **1412** having a value of 30 may be deducted from total score **1418**, and thus, the PEP may ultimately be classified as low-risk.

[0170] Referring next to FIG. **15**, FIG. **15** illustrates an example scoring table related to determining a false positive probability score used in an implementation. FIG. **15** shows table **1500**, which may include data parameters and weights assigned thereto for use by a scoring system (e.g., risk assessment platform **115**) when performing risk-based scoring techniques (e.g., method **201**, method **202**).

[0171] Table **1500** includes geography factors **1510** and corresponding points **1514**, commonness of last name factors **1511** and corresponding points **1514**, and missing factors **1512** and corresponding points **1514**. The factors and points **1514** may be used by a risk assessment platform (e.g., risk assessment platform **115**) to generate a false positive probability score, among other scores.

[0172] Geography factors **1510** may include various groups of countries that correspond to different amounts of points **1514**. For example, geography factors **1510** may include a first group of countries including China, India, the United States of America, and Canada. A PEP including an indication of one of these countries may be associated with a point value of 20.

[0173] Geography factors **1510** may include a second group of countries including Indonesia, Pakistan, Nigeria, Brazil, Bangladesh, Russia, and Mexico. A PEP including an indication of one of these countries may be associated with a point value of 10. Geography factors **1510** may include a third group including all other countries, which may be associated with a point value of 5.

[0174] Commonness of last name factors **1511** may include various groups of last names based on ethnicity that each correspond to a different amount of points **1514**. In various embodiments, a PEP including an indication of a last name in any of the groups identified in a common last name data set may be associated with a point value of 40. Examples of the groups within commonness of last name factors **1511** include 100 most common traditionally white names, 50 most common Asian names, 50 most common Hispanic names, and 30 most common Indian names. Other ethnic groups and numbers of last names may be contemplated.

[0175] Missing factors **1512** may include groups of factors that correspond to different amounts of points **1514**. For example, missing factors **1512** may include a first group including date of birth. For a PEP profile including an indication that the PEP profile is missing the date of birth of an individual, a scoring system may assign 25 points to a false positive probability score associated with the PEP profile. Missing factors **1512** may also include a second group including gender. For a PEP profile including an indication that the PEP profile is missing the gender of an individual, a scoring system may assign 25 points to the false positive probability score associated with the PEP profile.

[0176] Following analysis of the factors and records of a PEP profile, a scoring system may total the number of points **1514** associated with the PEP profile to generate a false positive probability score for the PEP profile. It follows that the higher the false positive probability score, the stronger the probability that a PEP profile is a false positive. The scoring system can compare the false positive probability score with a threshold scoring scale **1515**. In some embodiments, a score of 86 points or higher may be indicated as very high, a score between 70 points and 85 points may be indicated as high, and a score of 69 points and below may be indicated as lower than high or very high. The scoring system may provide indications of the false positive probability score per PEP profile to a user device (e.g., user device **120**) for display on a user interface (e.g., user interface **1100**) of the user device. Further, the scoring system may group PEP profiles into queues based on false positive probability scores.

[0177] In some embodiments, additional, fewer, or different factors, point values, and/or threshold scoring scales may be used by a scoring system in producing false positive probability scores associated with PEP profiles.

[0178] FIG. **16** illustrates computing system **1601** to perform risk assessment, risk analysis, threat level computation and categorization, and false positive probability computation and categorization, among other risk assessment and entity evaluation processes according to an implementation of the present technology. Computing system **1601** is representative of any system or collection of systems with which the various operational architectures, processes, scenarios, and sequences disclosed herein for risk assessment and related operations may be employed. Computing system **1601** may be implemented as a single apparatus, system, or device or may be implemented in a distributed manner as multiple apparatuses, systems, or devices. Computing system **1601** includes, but is not limited to, processing system **1602**, storage system **1603**, software **1605**, communication interface system **1607**, and user interface system **1609** (optional). Processing system **1602** is operatively coupled with storage system **1603**, communication interface system **1607**, and user interface system **1609**. Computing system **1601** may be representative of a cloud computing device, distributed computing device, or the like.

[0179] Processing system **1602** loads and executes software **1605** from storage system **1603**. Software **1605** includes and implements risk assessment process **1606**, which is representative of any of the data analyzing, sorting, evaluating, normalizing, standardizing, weighting, scoring, categorization, configuring, and visualizing processes discussed with respect to the preceding Figures. When executed by processing system **1602** to provide risk assessment functions, software **1605** directs processing system **1602** to operate as described herein for at least the various processes, operational scenarios, and sequences discussed in the foregoing implementations. Computing system **1601** may optionally include additional devices, features, or functionality not discussed for purposes of brevity.

[0180] Referring still to FIG. **16**, processing system **1602** may comprise a micro-processor and other circuitry that retrieves and executes software **1605** from storage system **1603**. Processing system **1602** may be implemented within a single processing device but may also be distributed across multiple processing devices or sub-systems that cooperate in executing program instructions. Examples of processing system **1602** include general purpose central processing units, graphical processing units, application specific processors, and logic devices, as well as other processing devices and combinations or variations thereof.

[0181] Storage system **1603** may comprise any computer readable storage media readable by processing system **1602** and capable of storing software **1605**. Storage system **1603** may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Examples of storage media include random access memory, read only memory, magnetic disks, optical disks, optical media, flash memory, virtual memory and non-virtual memory, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other suitable storage media.

[0182] In no case is the computer readable storage media a propagated signal.

[0183] In addition to computer readable storage media, in some implementations storage system **1603** may also include computer readable communication media over which at least some of software **1605** may be communicated internally or externally. Storage system **1603** may be implemented as a single storage device but may also be implemented across multiple storage devices or sub-systems co-located or distributed relative to each other. Storage system **1603** may comprise additional elements, such as a controller, capable of communicating with processing system **1602** or possibly other systems.

[0184] Software **1605** (including risk assessment process **1606**) may be implemented in program instructions and among other functions may, when executed by processing system **1602**, direct processing system **1602** to operate as described with respect to the various operational scenarios, sequences, and processes illustrated herein. For example, software **1605** may include program instructions for implementing risk assessment processes as described herein.

[0185] In particular, the program instructions may include various components or modules that cooperate or otherwise interact to carry out the various processes and operational scenarios described herein. The various components or modules may be embodied in compiled or interpreted instructions, or in some other variation or combination of instructions. The various components or modules may be executed in a synchronous or asynchronous manner, serially or in parallel, in a single threaded environment or multi-threaded, or in accordance with any other suitable execution paradigm, variation, or combination thereof. Software **1605** may include additional processes, programs, or components, such as operating system software, virtualization software, or other application software. Software **1605** may also comprise firmware or some other form of machine-readable processing instructions executable by processing system **1602**.

[0186] In general, software **1605** may, when loaded into processing system **1602** and executed, transform a suitable apparatus, system, or device (of which computing system **1601** is representative) overall from a general-purpose computing system into a special-purpose computing system customized to provide memory allocation as described herein. Indeed, encoding software **1605** on storage system **1603** may transform the physical structure of storage system **1603**. The specific transformation of the physical structure may depend on various factors in different implementations of this description. Examples of such factors may include, but are not limited to, the technology used to implement the storage media of storage system **1603** and whether the computer-storage media are characterized as primary or secondary storage, as well as other factors.

[0187] For example, if the computer readable storage media are implemented as semiconductor-based memory, software **1605** may transform the physical state of the semiconductor memory when the program instructions are encoded therein, such as by transforming the state of transistors, capacitors, or other discrete circuit elements constituting the semiconductor memory. A similar transformation may occur with respect to magnetic or optical media. Other transformations of physical media are possible without departing from the scope of the present description, with the foregoing examples provided only to facilitate the present discussion.

[0188] Communication interface system **1607** may include communication connections and devices that allow for communication with other computing systems (not shown) over communication networks (not shown). Examples of connections and devices that together allow for inter-system communication may include network interface cards, antennas, power amplifiers, radiofrequency circuitry, transceivers, and other communication circuitry. The connections and devices may communicate over communication media to exchange communications with other computing systems or networks of systems, such as metal, glass, air, or any other suitable communication media. The aforementioned media, connections, and devices are well known and need not be discussed at length here.

[0189] Communication between computing system **1601** and other computing systems (not shown), may occur over a communication network or networks and in accordance with various communication protocols, combinations of protocols, or variations thereof. Examples include intranets, internets, the Internet, local area networks, wide area networks, wireless networks, wired networks, virtual networks, software defined networks, data center buses and backplanes, or any other type of network, combination of networks, or variation thereof. The aforementioned communication networks and protocols are well known and need not be discussed at length here.

[0190] While some examples provided herein are described in the context of a risk assessment platform, system, apparatus, device, environment, server, or service, the risk assessment and analysis methods, techniques, systems, and devices described herein are not limited to such examples and may apply to a variety of other processes, systems, applications, devices, and the like. As will be appreciated by one skilled in the art, aspects of the present invention may be embodied as a system, method, computer program product, and other configurable systems. Accordingly, aspects of the present invention may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a "circuit," "module" or "system." Furthermore, aspects of the present invention may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

[0191] Unless the context clearly requires otherwise, throughout the description and the claims, the words "comprise," "comprising," and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of "including, but not limited to." As used herein, the terms "connected," "coupled," or any variant thereof means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words "herein," "above," "below," and words of similar import, when used in this application, refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word "or," in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

[0192] The phrases "in some examples," "according to some examples," "in the examples shown," "in other examples," and the like generally mean the particular feature, structure, or characteristic following the phrase is included in at least one implementation of the present technology, and may be included in more than one implementation. In addition, such phrases do not necessarily refer to the same example or different examples.

[0193] The above Detailed Description of examples of the technology is not intended to be exhaustive or to limit the technology to the precise form disclosed above. While specific examples for the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative implementations may perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or subcombinations. Each of these processes or blocks may be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks may instead be performed or implemented in parallel or may be performed at different times. Further any specific numbers noted herein are only examples: alternative implementations may employ differing values or ranges.

[0194] The teachings of the technology provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various examples described above can be combined to provide further implementations of the technology.

[0195] Some alternative implementations of the technology may include not only additional elements to those implementations noted above, but also may include fewer elements.

[0196] These and other changes can be made to the technology in light of the above Detailed Description. While the above description describes certain examples of the technology, and describes the best mode contemplated, no matter how detailed the above appears in text, the technology can be practiced in many ways. Details of the system may vary considerably in its specific implementation, while still being encompassed by the technology disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the technology should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the technology with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the technology to the specific examples disclosed in the specification, unless the above Detailed Description section explicitly defines such terms. Accordingly, the actual scope of the technology encompasses not only the disclosed examples, but also equivalent ways of practicing or implementing the technology under the claims.

[0197] To reduce the number of claims, certain aspects of the technology are presented below in certain claim forms, but the applicant contemplates the various aspects of the technology in any number of claim forms. For example, while only one aspect of the technology is recited as a computer-readable medium claim, other aspects may likewise be embodied as a computer-readable medium claim, or in other forms, such as being embodied in a means-plus-function claim. Any claims intended to be treated under 35 U.S.C. § 112(f) will begin with the words "means for" but use of the term "for" in any other context is not intended to invoke treatment under 35 U.S.C. § 112(f). Accordingly, the applicant reserves the right to pursue additional claims after filing this application to pursue such additional claim forms, in either this application or in a continuing application.

What is claimed is:

1. A risk assessment platform comprising:

one or more computer-readable storage media;

program instructions stored on the one or more computer-readable storage media that, based on being read and executed by a processing device, direct the processing device to:

receive an input comprising an indication of a target entity subject to a risk assessment;

identify, based on the input, multiple possible instances of the target entity from a database including a plurality of entities;

for each possible instance of the multiple possible instances, evaluate a strength of a match of the possible instance relative to the target entity based on applying a set of criteria against the indication of the target entity, applying the set of criteria against an indication of the possible instance, and comparing results of the applications of the set of criteria;

display, on a graphical user interface, a visual representation of the multiple possible instances and corresponding strengths of matches; and

in response to a selection of one or more of the multiple possible instances, add the one or more of the multiple possible instances to a queue associated with a further risk assessment.

2. The risk assessment platform of claim **1**, wherein the indication of the target entity comprises personal information and geographical information associated with the target entity.

3. The risk assessment platform of claim **2**, wherein:

the set of criteria includes sets of weights applicable to the personal information and the geographical information associated with the target entity;

a first set of weights of the sets of weights corresponds to a commonness of a surname identified in the indications of the target entity and the multiple possible instances; and

a second set of weights of the sets of weights corresponds to a popularity of a domicile identified in the indications of the target entity and the multiple possible instances.

4. The risk assessment platform of claim **3**, wherein to apply the set of criteria against the indication of the target entity, the program instructions direct the processing device to:

access a first database and obtain common surname data from the first database, wherein the common surname data includes lists of common surnames for multiple ethic groups;

determine the first set of weights for the common surname data;

access a second database and obtain popular country data from the second database;

determine the second set of weights for the popular country data; and

generate the strengths of matches for the multiple possible instances of the target entity based on applying the first and second sets of weights to surnames and domiciles derived from the indications of the target entity and the multiple possible instances.

5. The risk assessment platform of claim **4**, wherein, for each possible instance, evaluating the strength of a match for the possible instance comprises identifying a combination of a missing date of birth and a missing gender in the indication of the possible instance and applying a weight based on the combination to the results of the application of the criteria.

6. The risk assessment platform of claim **1**, wherein the program instructions further direct the processing device to, in response to a dismissal input of one or more other possible instances of the multiple possible instances, updating metadata associated with the one or more other possible instances to indicate the one or more other possible instances as false positive instances with respect to the target entity.

7. A risk assessment platform comprising:

one or more computer-readable storage media;

program instructions stored on the one or more computer-readable storage media that, based on being read and executed by a processing device, direct the processing device to:

receive an input comprising an indication of a target entity subject to a risk assessment;

identify, based on the input, multiple possible instances of the target entity from a database including a plurality of entities;

display, on a graphical user interface, a visual representation of the multiple possible instances, corresponding strengths of matches relative to the target entity, and corresponding risk profiles; and

in response to a selection of a risk profile of one of the multiple possible instances, display, on the graphical user interface, a visual representation of a risk assessment generated for the risk profile based on performing a risk analysis on data associated with the risk profile accessed through one or more databases, wherein the visual representation of the risk assessment includes an indication of a threat confidence level associated with the risk profile.

8. The risk assessment platform of claim **7**, wherein the indication of the target entity comprises personal information and geographical information associated with the target entity.

9. The risk assessment platform of claim **7**, wherein the program instructions direct the processing device to determine the strengths of matches of the multiple possible instances relative to the target entity based on applying a set of criteria against the indication of the target entity and comparing results of the application of the set of criteria with the indications of the multiple possible instances.

10. The risk assessment platform of claim **7**, wherein the data associated with the risk profile comprises corruption perception index data and politically-exposed persons data.

11. The risk assessment platform of claim **10**, wherein to perform the risk analysis on the data, the program instructions direct the processing device to:

identify a subset of records of the data having associated values;

convert the values associated with the subset of records from a first format to a second format; and

generate sets of weights for the data, wherein the sets of weights comprises a first set of weights including the converted values, and a second set of weights; and

apply the sets of weights to the data.

12. The risk assessment platform of claim **7**, wherein the threat confidence level corresponds to a level of confidence of corruption with respect to performing financial transactions with the entity.

13. The risk assessment platform of claim **12**, wherein the program instructions direct the processing device to output an alert indication based on determining that the threat confidence level exceeds a threshold level.

14. The risk assessment platform of claim **7**, wherein the program instructions direct the processing device to, in response to a confirmation input of a risk profile, add the risk profile to a queue associated with risk profiles subject to further risk assessment.

15. A risk assessment system comprising:

a memory; and

a processor coupled with executable instructions forming modules of the risk assessment system and configured to execute the modules of the risk assessment system to produce a risk assessment, wherein the modules executed by the processor include:

an input interface configured to obtain, from one or more databases, risk and governance data associated with an entity subject to the risk assessment, and obtain, from a user device, an input comprising an indication of a target entity;

an output interface configured to output the risk assessment of the target entity;

a data processing module configured to:

receive the risk and governance data associated with the target entity from the input interface and identify a subset of records having associated values;

convert the values associated with the subset of records of the risk and governance data from a first format to a second format; and

provide the converted values to a weight generation module;

the weight generation module configured to:

receive the risk and governance data and the converted values from the data processing module;

generate a set of weights for the risk and governance data, wherein the set of weights comprises a first subset of weights including the converted values, and a second subset of weights; and

provide the set of weights to a scoring module; and

the threat detection module configured to:

generate the risk assessment of the target entity based on applying the set of weights to the risk and governance data; and

provide the risk assessment to the output interface.

16. The risk assessment system of claim **15**, wherein the risk assessment includes a threat confidence level associated with the risk profile, and wherein the threat confidence level corresponds to a level of confidence of corruption with respect to performing financial transactions with the entity.

17. The risk assessment system of claim **16**, wherein the output interface is further configured to output an alert indication responsive to determining that the threat confidence level exceeds a threshold level.

18. The risk assessment system of claim **16**, wherein the modules executed by the processor further include a visualization module configured to receive the risk assessment and display, on a graphical user interface, a visual representation of a risk assessment including a visual indication of the threat confidence level.

19. The risk assessment system of claim **15**, wherein the modules executed by the processor further include a false positive analyzer module configured to:

receive the risk and governance data associated with the entity from the input interface;

identify multiple possible instances of the target entity from the risk and governance data;

for each possible instance of the multiple possible instances, evaluate a strength of a match of the possible instance to the target entity based on applying a set of criteria against the indication of the target entity and comparing results of the application of the set of criteria with an indication of the possible instance; and

for each possible instance of the multiple possible instances having a strength of a match below a threshold level, update metadata of the possible instance to indicate the possible instance as a false positive instance with respect to the target entity.

20. The risk assessment system of claim **19**, wherein the modules executed by the processor further include a visualization module configured to receive indications of the strengths of matches of the multiple possible instances and display, on a graphical user interface, indications of the multiple possible instances and the indications of the strengths of matches.

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